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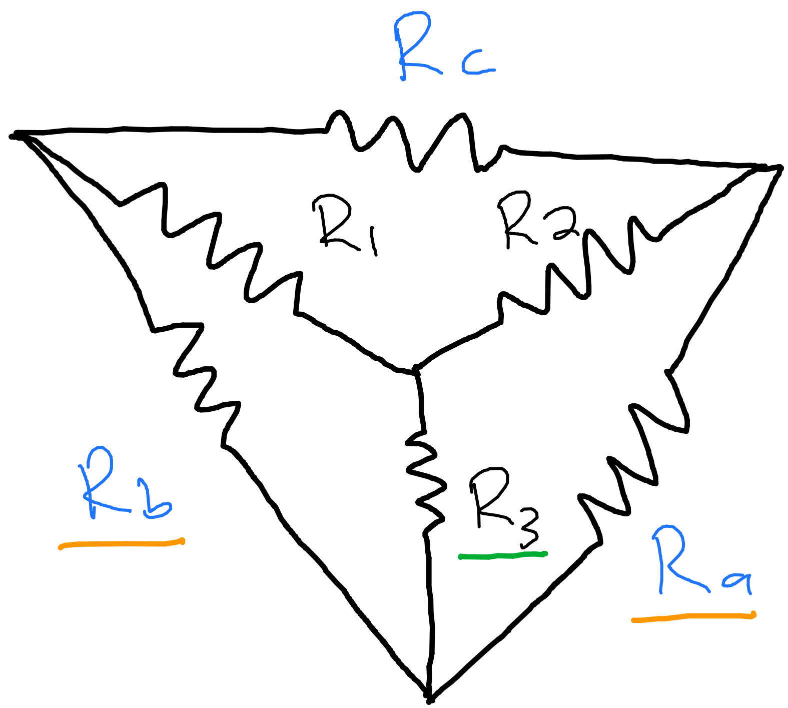
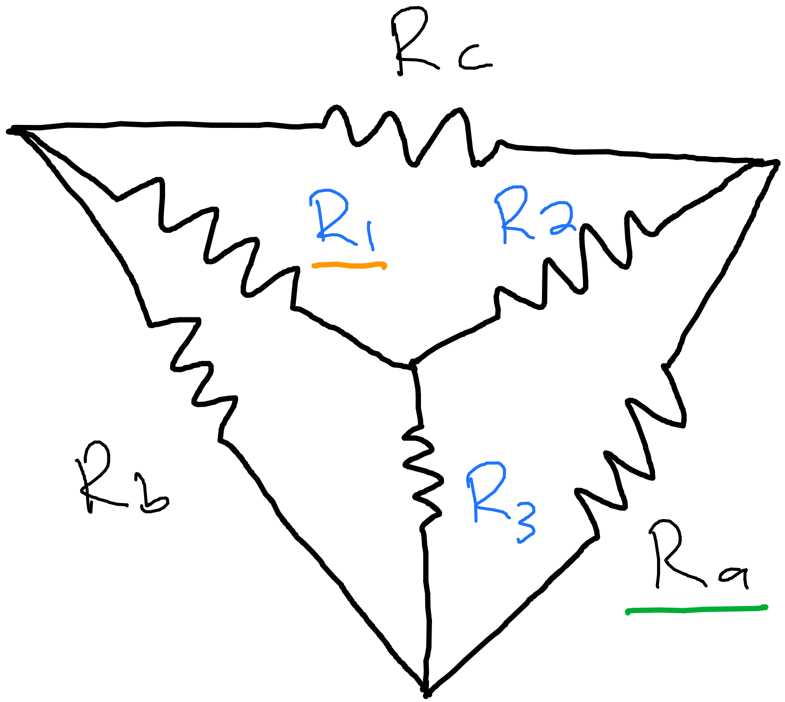
EBN111/122

Tutorial 3 – Memo
4 August 2017



$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

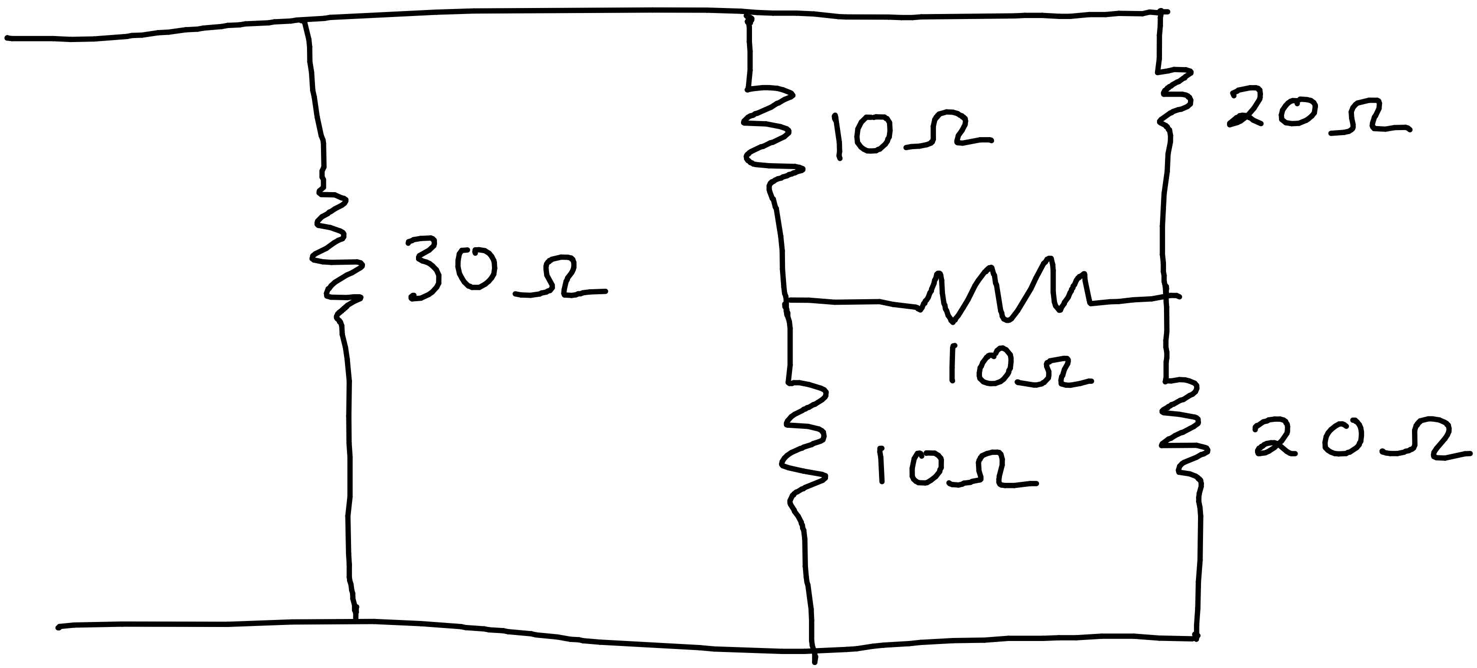
Each resistor in the Y network is the product of the resistors in the two adjacent Δ branches divided by the sum of the three Δ resistors.

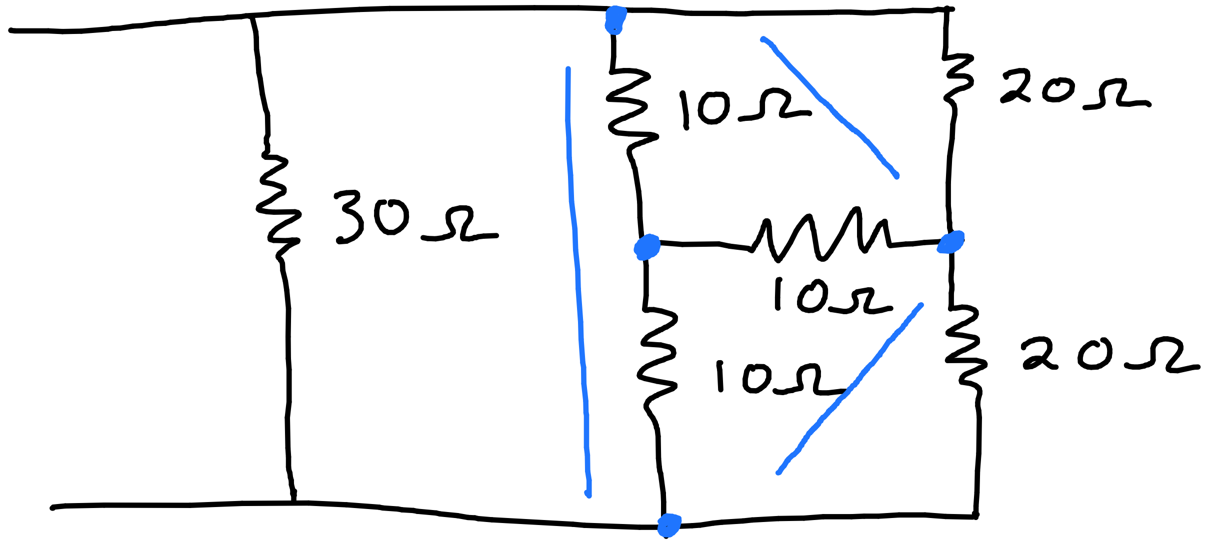


Each resistor in the Δ network is the sum of all possible products of Y resistor taken two at a time, divided by the opposite Y resistor.

$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}$$

2.51 a) Obtain the equivalent resistance.



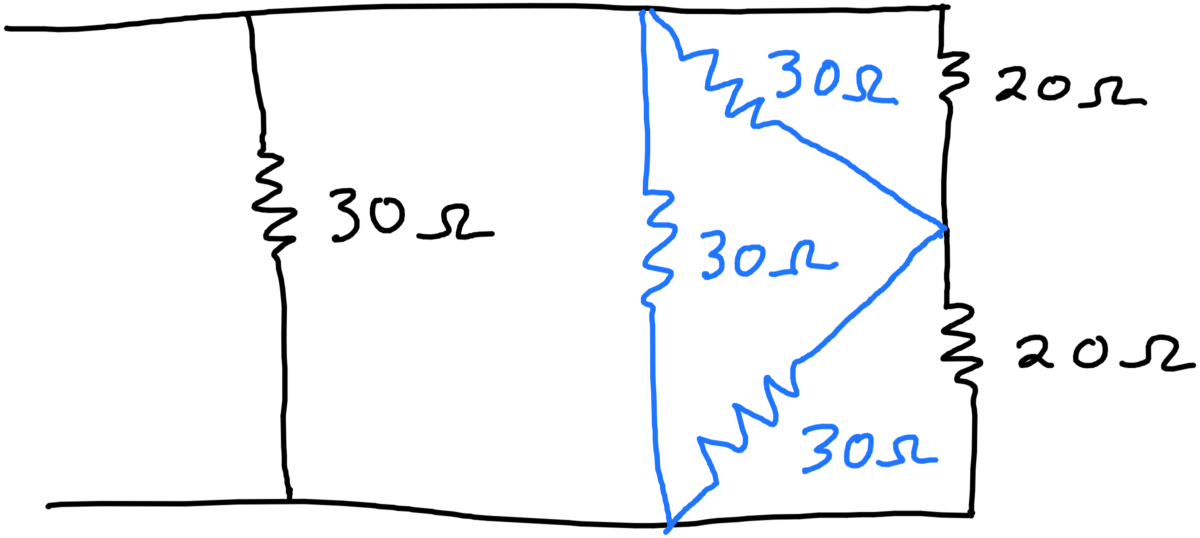


$Y \rightarrow \Delta$ special case

$$R_\Delta = 3R_Y$$

$$R_\Delta = 3 \times 10$$

$$= 30\,\Omega$$

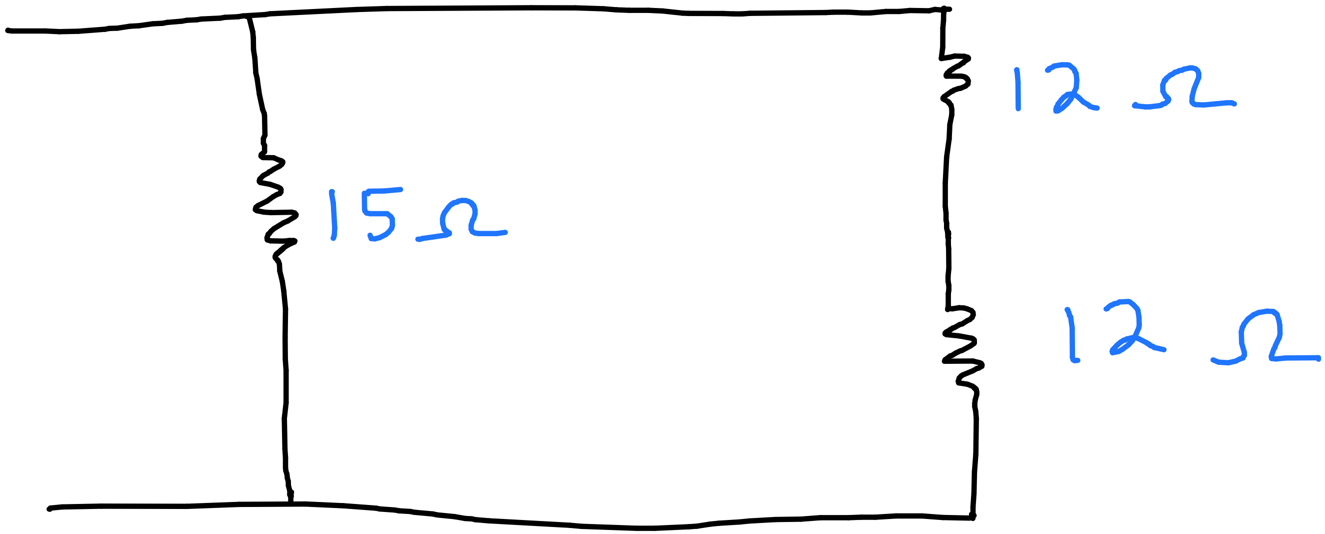


$$30 \| 30 = \left(\frac{1}{30} + \frac{1}{30}\right)^{-1}$$

$$= 15$$

$$30 \| 20 = \left(\frac{1}{20} + \frac{1}{30}\right)^{-1}$$

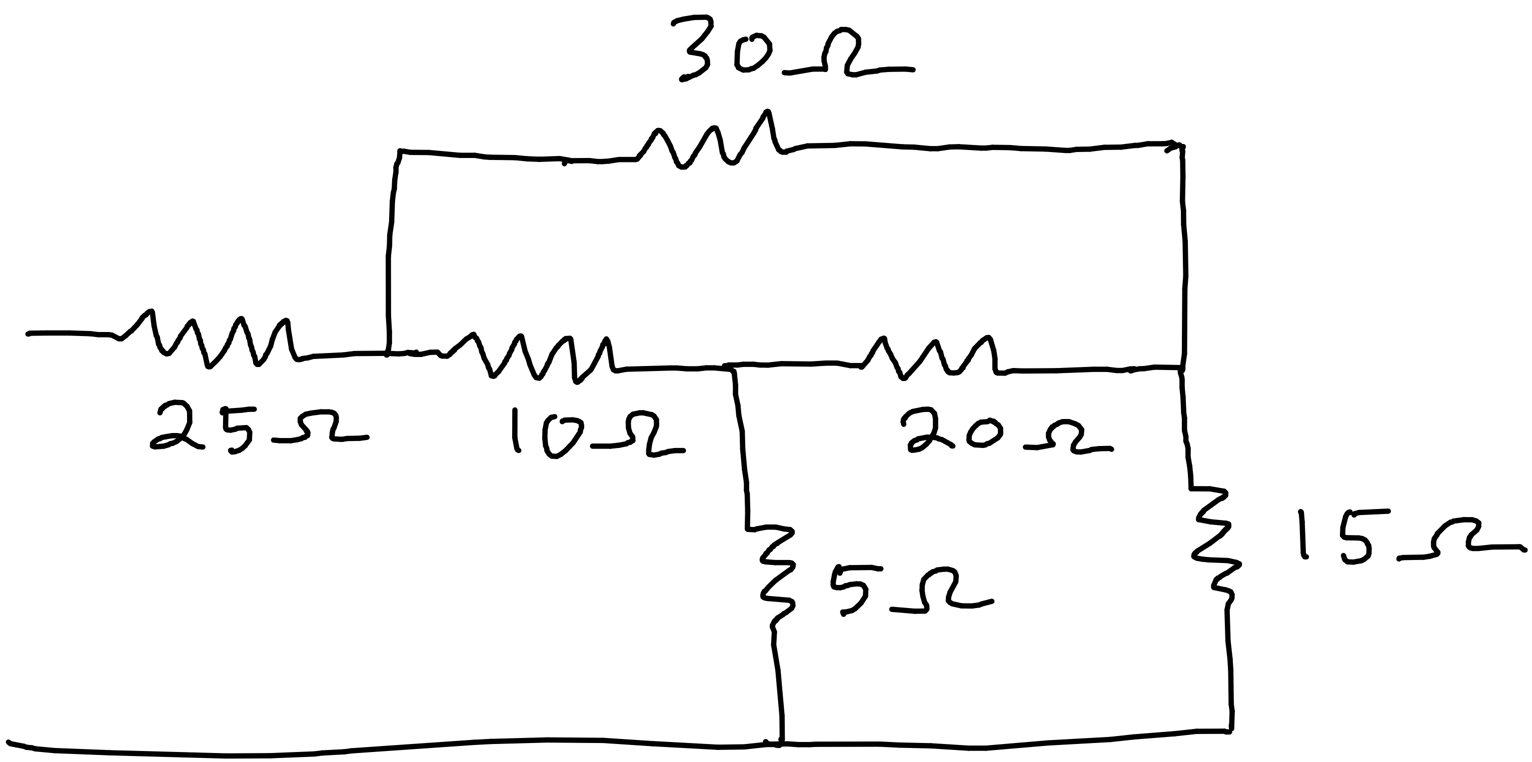
$$= 12\ \Omega$$

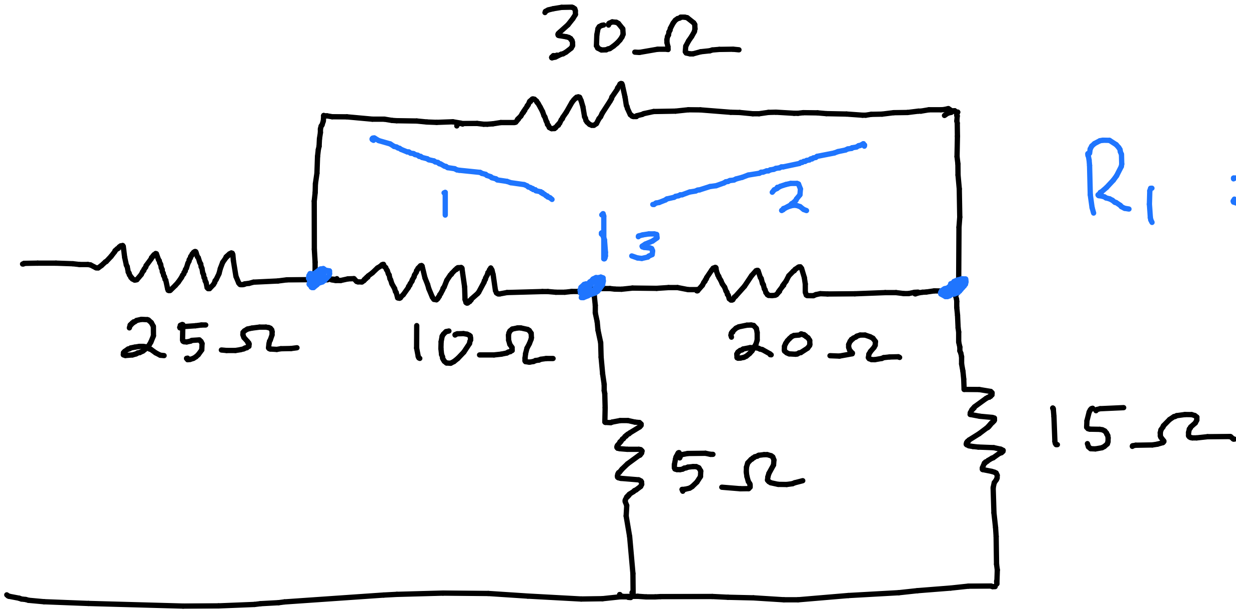


$$15 \,||\, (12+12) = \left(\frac{1}{15} + \frac{1}{12+12}\right)^{-1}$$

$$R_{eq} = 9.2308\ \Omega$$

2.51 b) Obtain the equivalent resistance



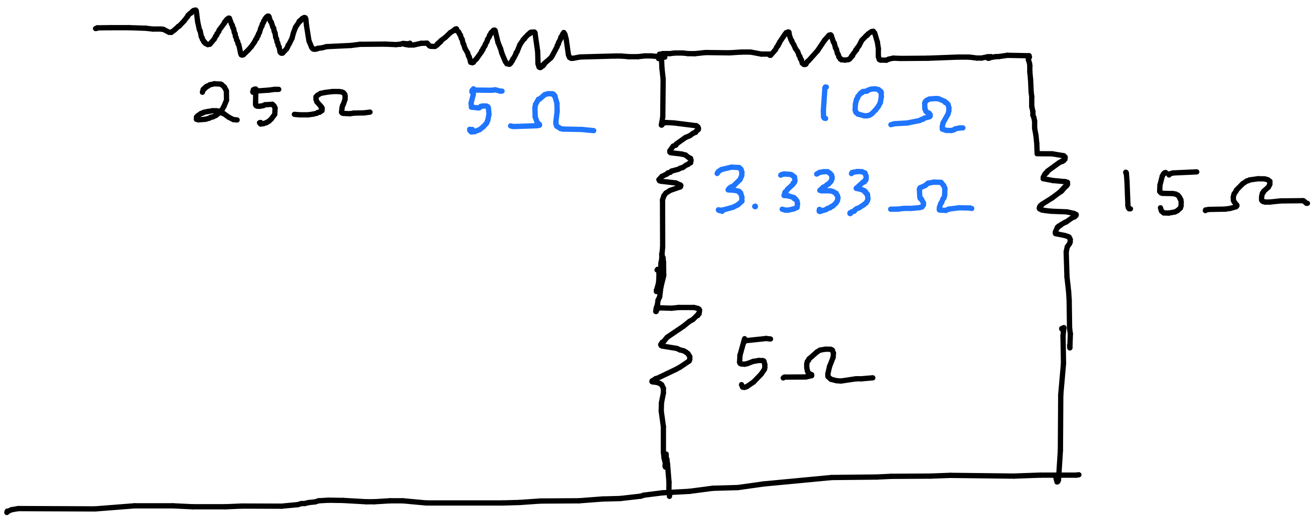


$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$\Delta - Y$ conversion

$$R_a + R_b + R_c = 20 + 10 + 30 = 60\,\Omega$$

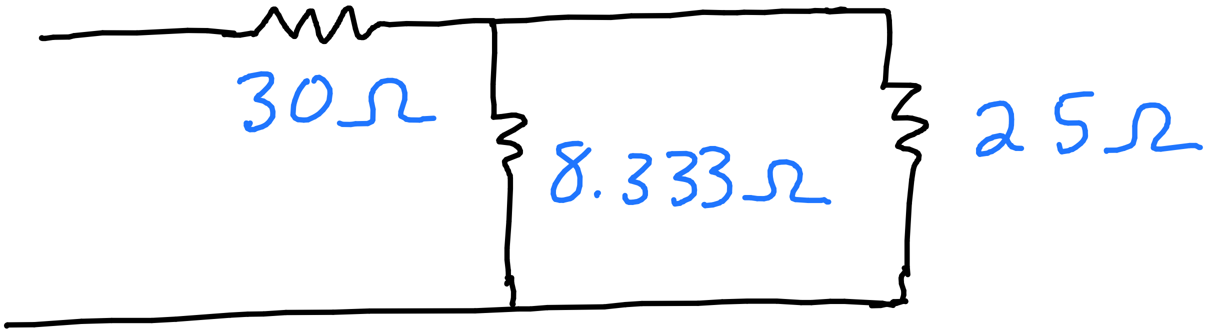
$$R_1 = \frac{10 \times 30}{60} = 5\,\Omega \qquad R_2 = \frac{20 \times 30}{60} = 10\,\Omega \qquad R_3 = \frac{10 \times 20}{60} = 3.333\,\Omega$$



$$25 + 5 = 30\ \Omega$$

$$3.333 + 5 = 8.333\ \Omega$$

$$10 + 15 = 25\ \Omega$$

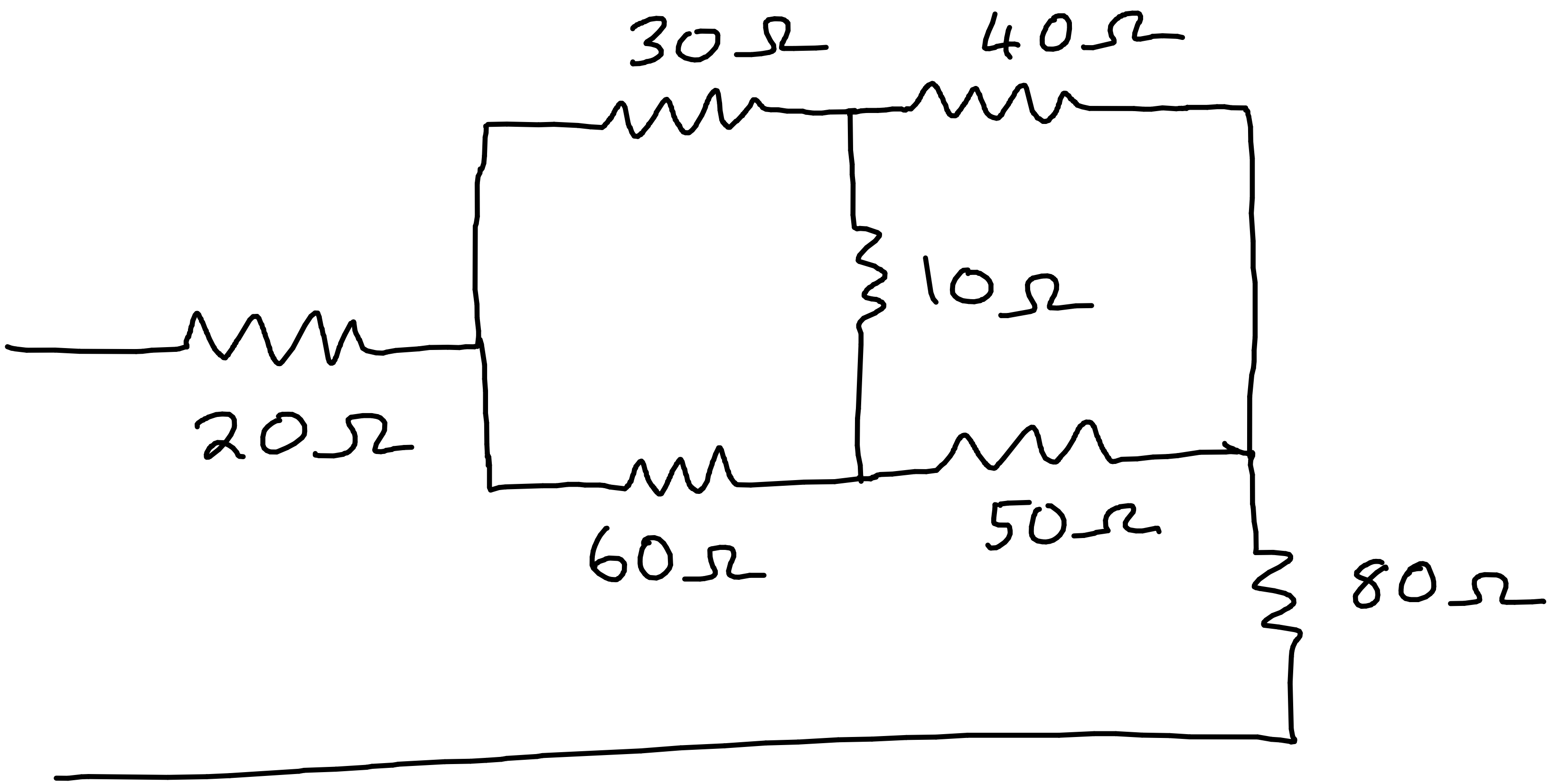


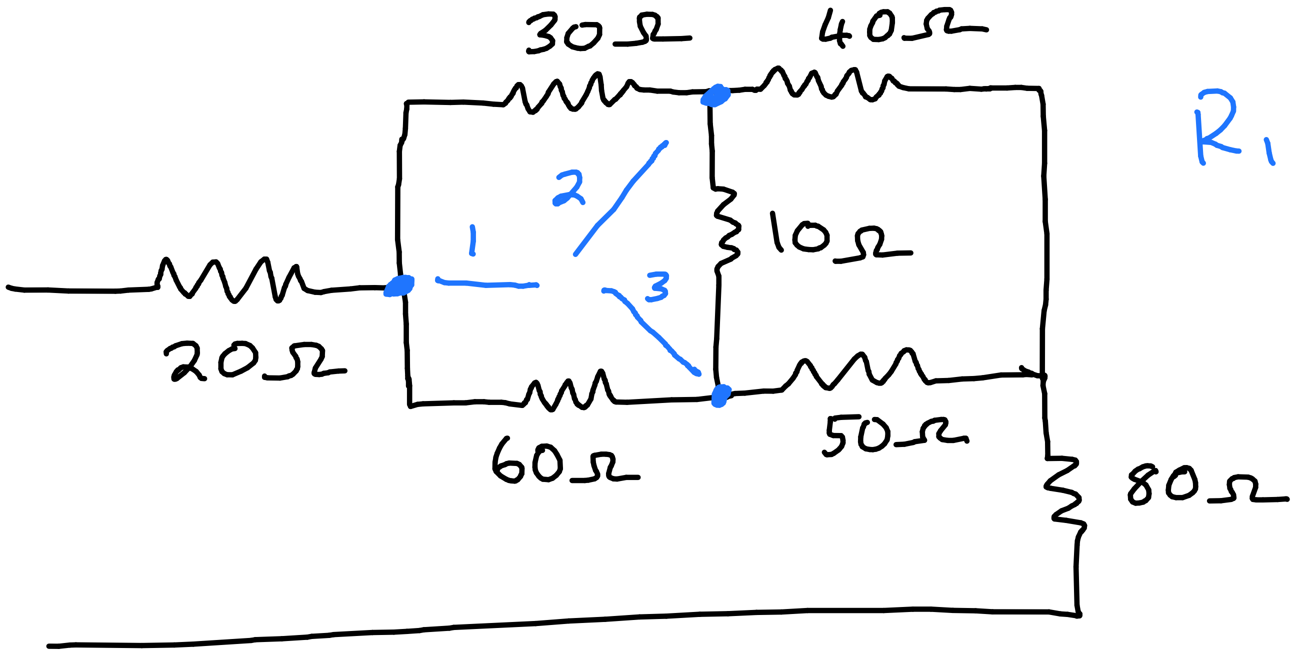
$$R_{eq} = 30 + 8.333 \| 25$$

$$= 30 + \left(\frac{1}{8.333} + \frac{1}{25}\right)^{-1}$$

$$= 36.250$$

2.53 a) Obtain the equivalent resistance





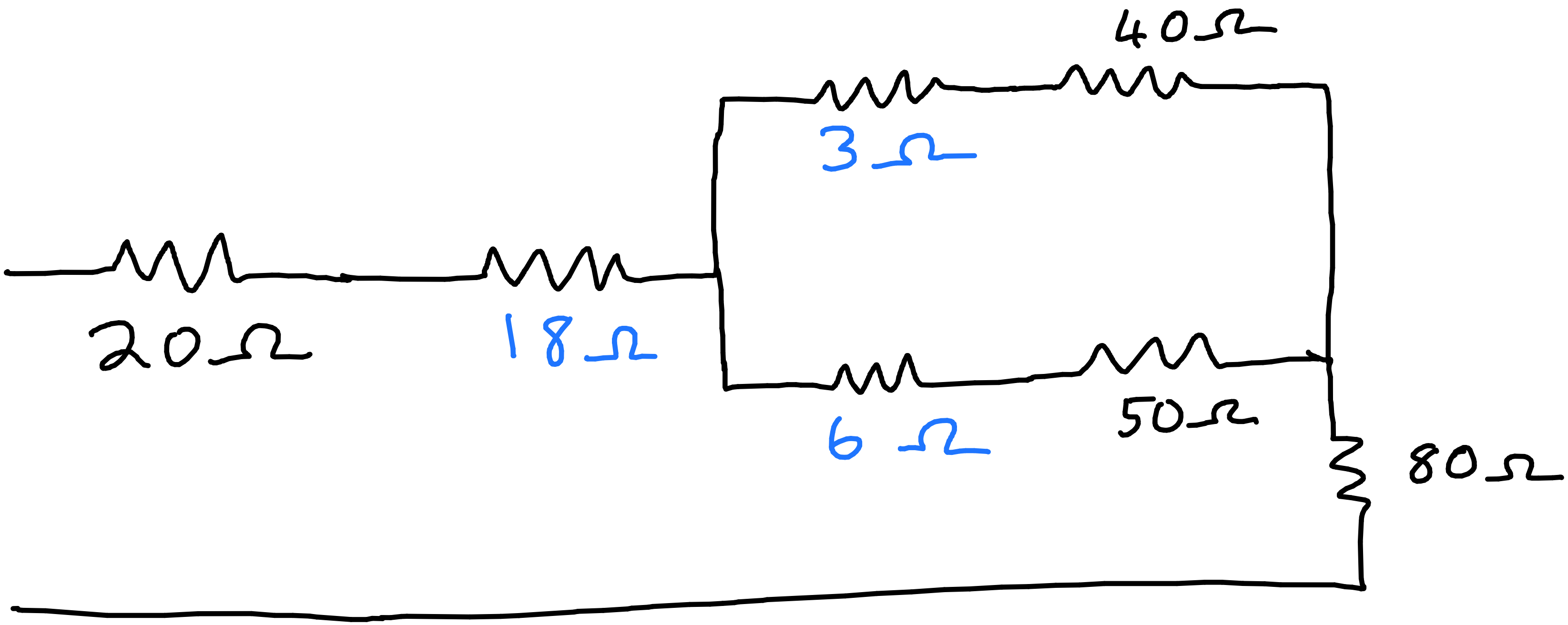
$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$\Delta - Y$ conversion

$$R_a + R_b + R_c = 10 + 30 + 60 = 100\,\Omega$$

$$R_1 = \frac{30 \times 60}{100} \qquad R_2 = \frac{30 \times 10}{100} \qquad R_3 = \frac{10 \times 60}{100}$$

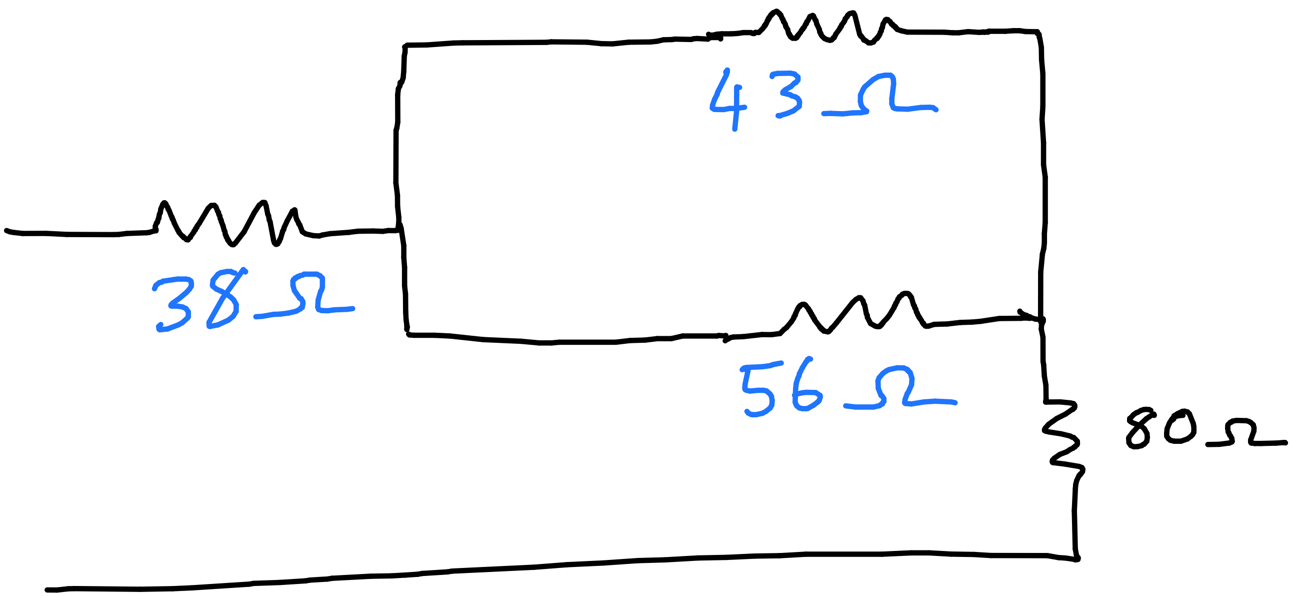
$$= 18\,\Omega \qquad = 3\,\Omega \qquad = 6\,\Omega$$



$20 + 18 = 38\,\Omega$

$3 + 40 = 43\,\Omega$

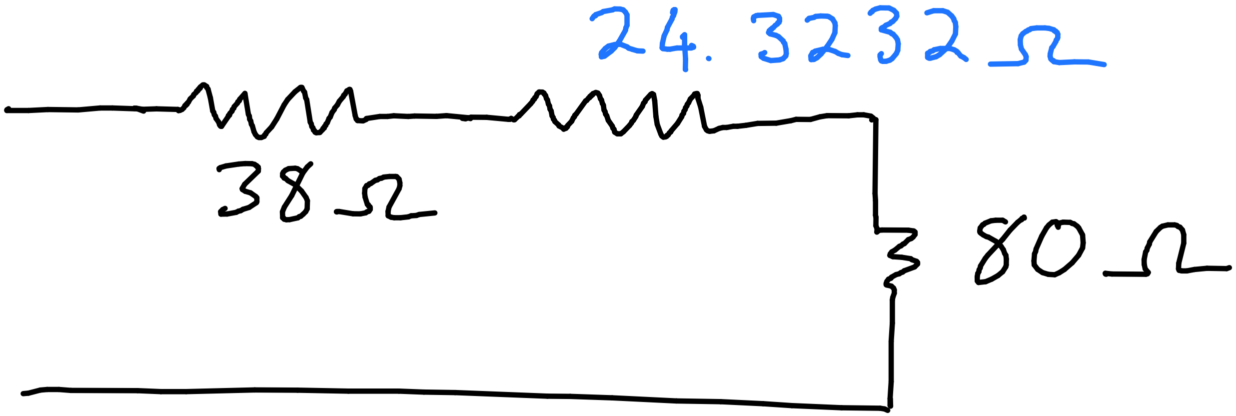
$6 + 50 = 56\,\Omega$



$$43 \parallel 56 = \left(\frac{1}{43} + \frac{1}{56}\right)^{-1}$$

$$= \frac{43 \times 56}{43 + 56}$$

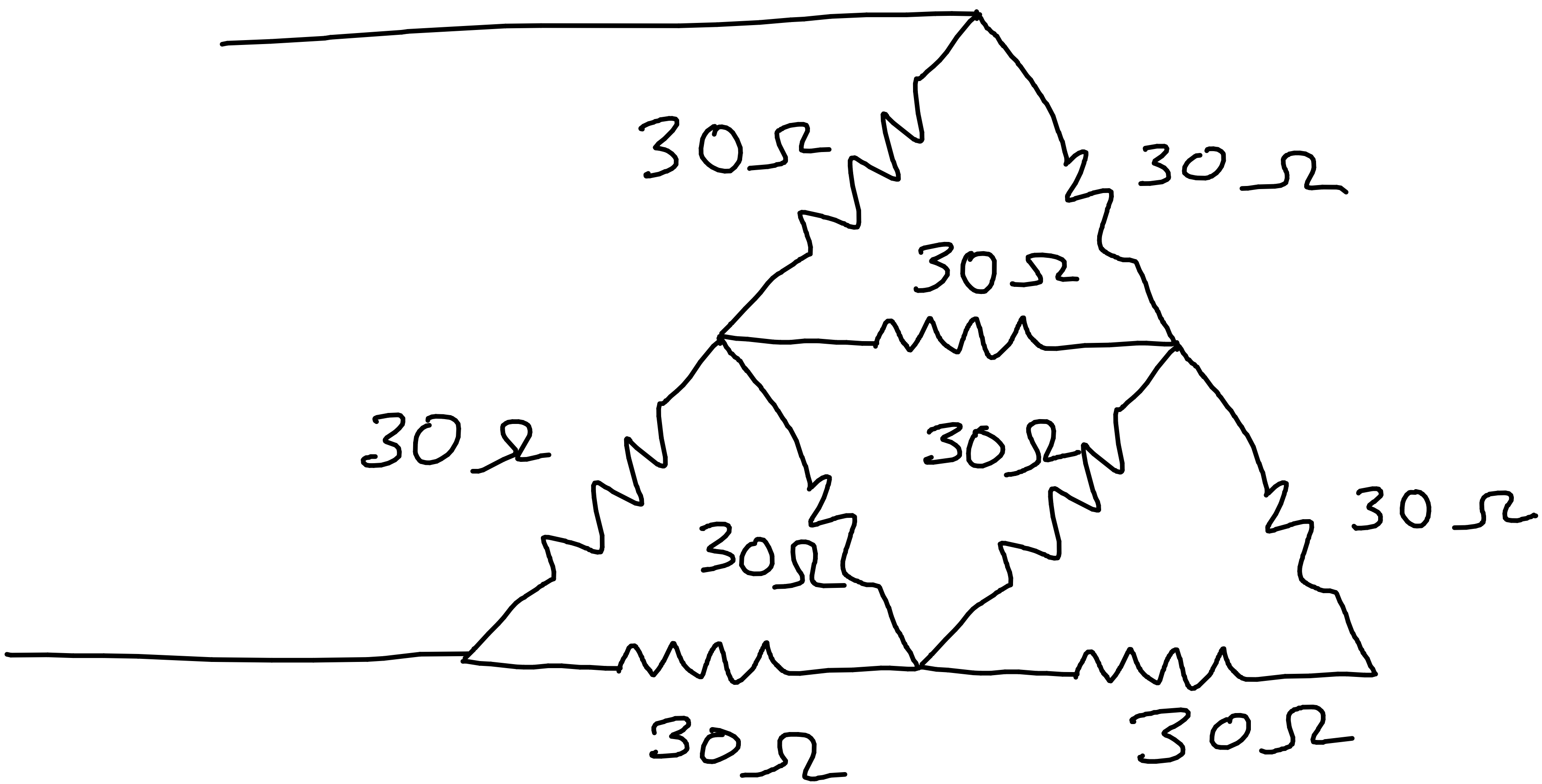
$$= 24.3232$$

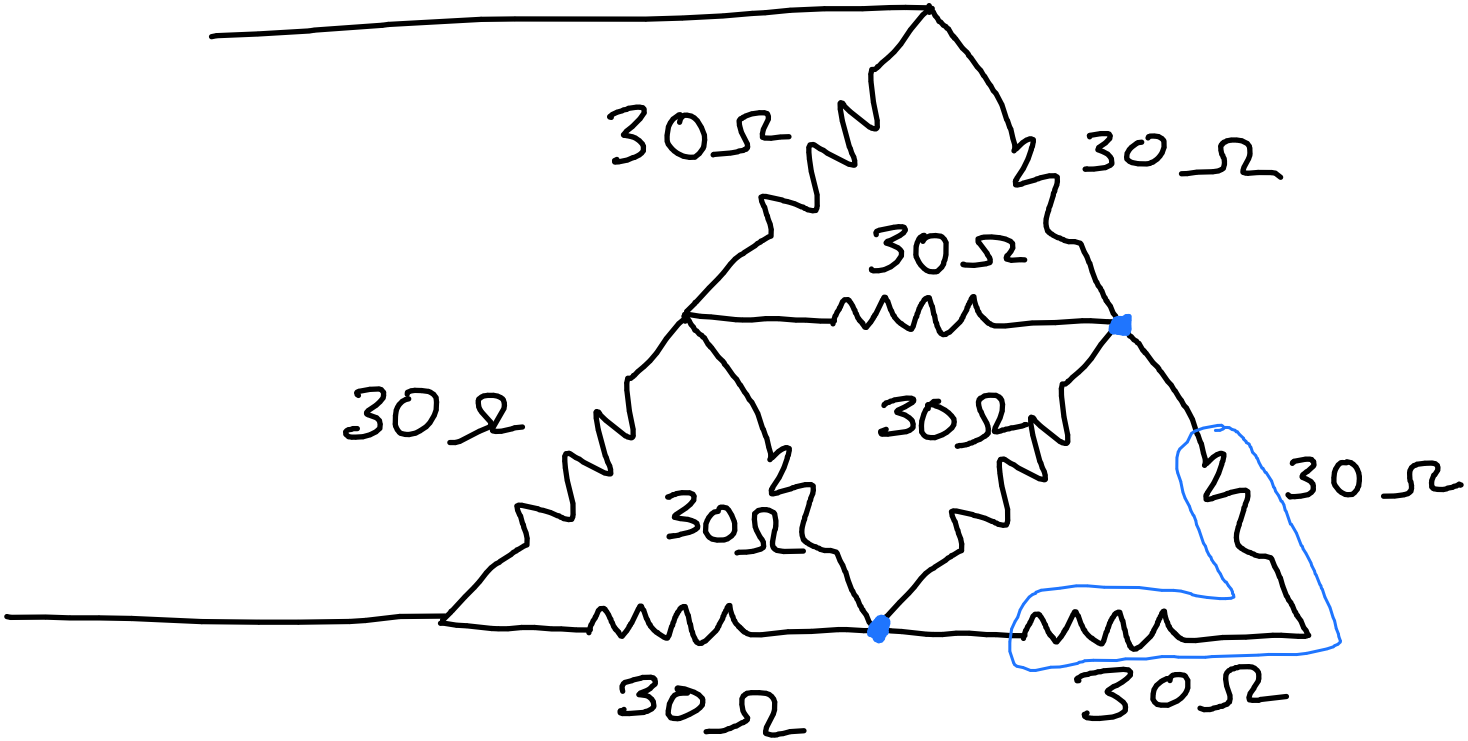


$$R_{eq} = 38 + 24.3232 + 80$$

$$= 142.3232\ \Omega$$

2.53 b) Obtain the equivalent resistance

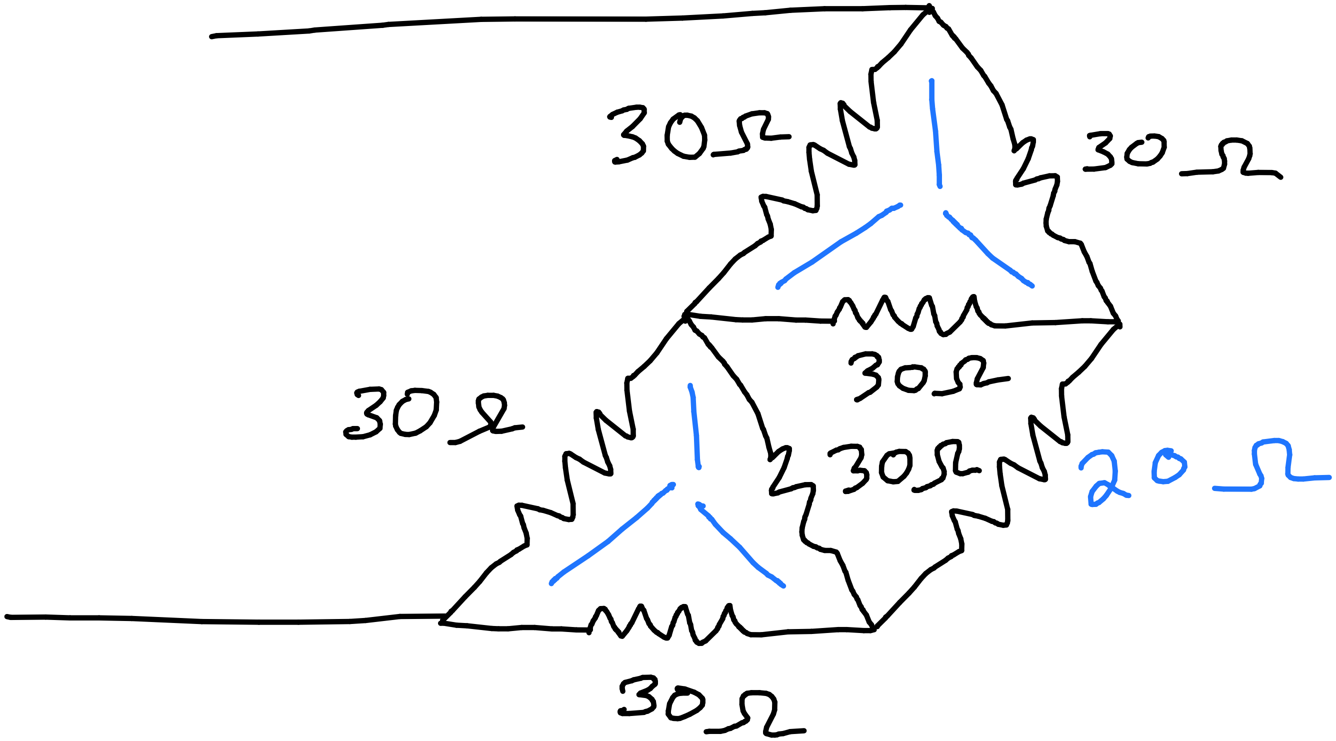




$$30 + 30 = 60\,\Omega$$

$$30 \parallel 60 = \frac{30 \times 60}{30 + 60}$$

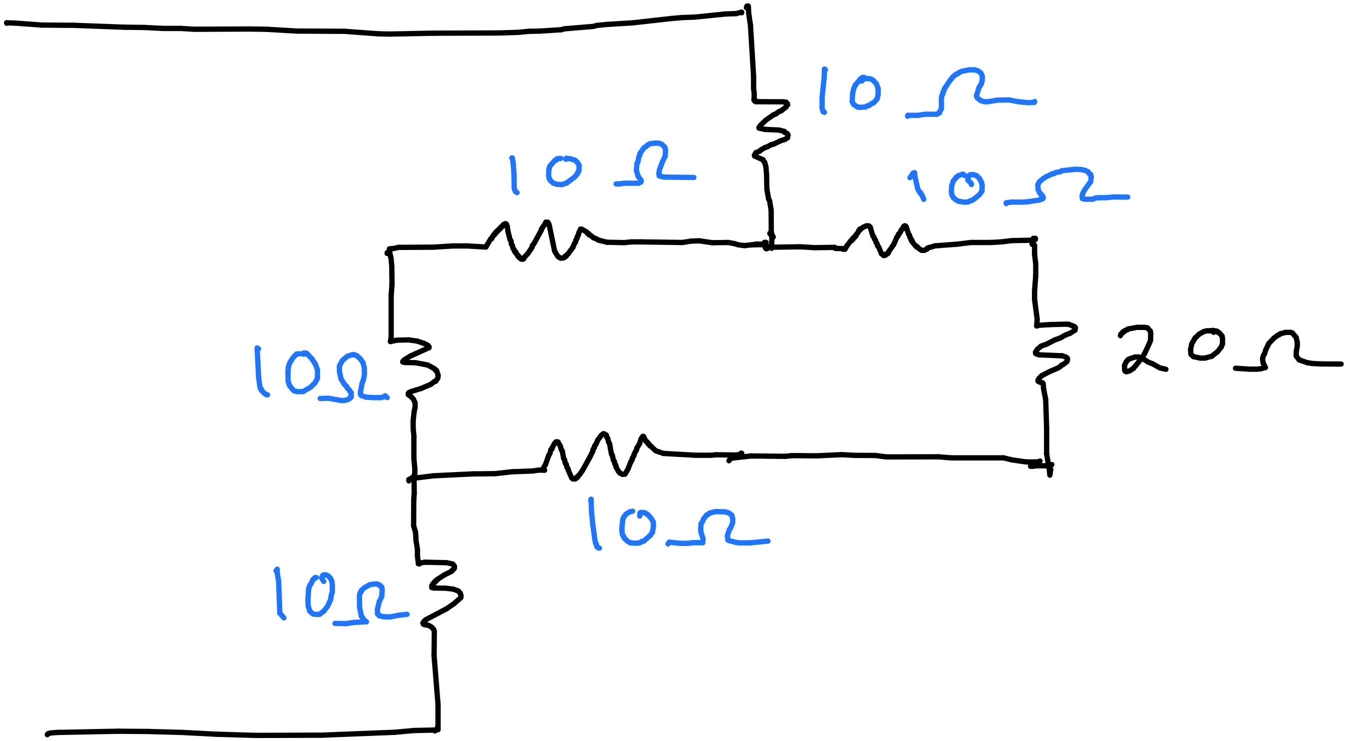
$$= 20\,\Omega$$



Δ - Y conversion special case

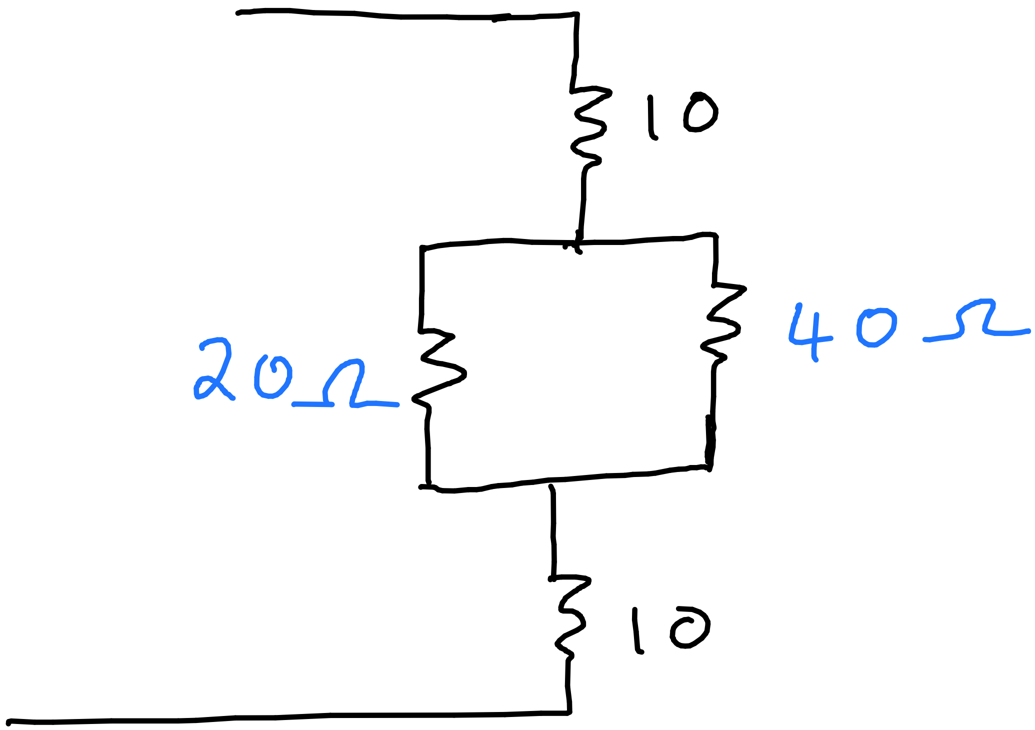
$$R_Y = \frac{R_\Delta}{3}$$

$$= \frac{30}{3} = 10\,\Omega$$



$$10 + 20 + 10 = 40\,\Omega$$

$$10 + 10 = 20\,\Omega$$

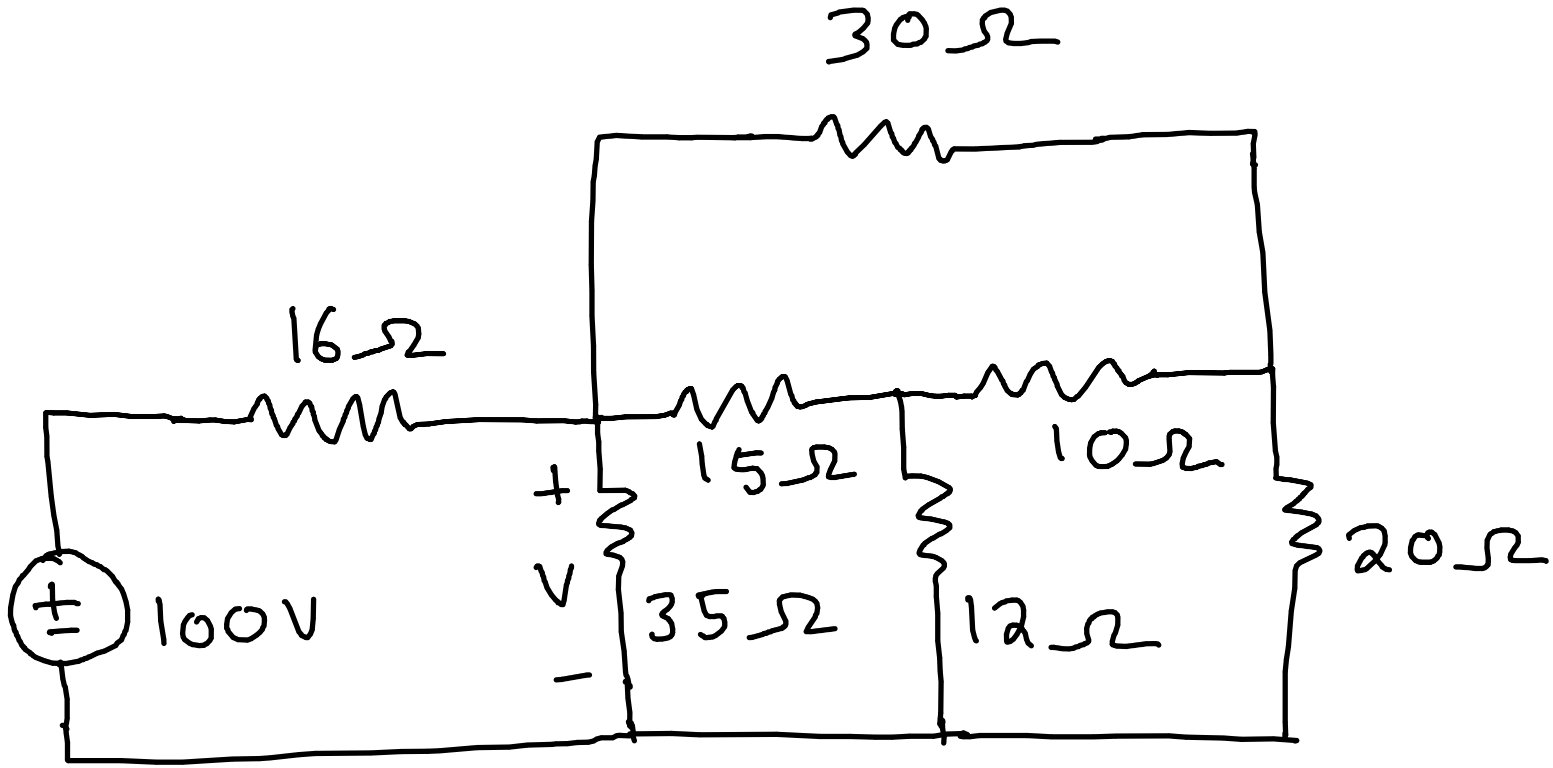


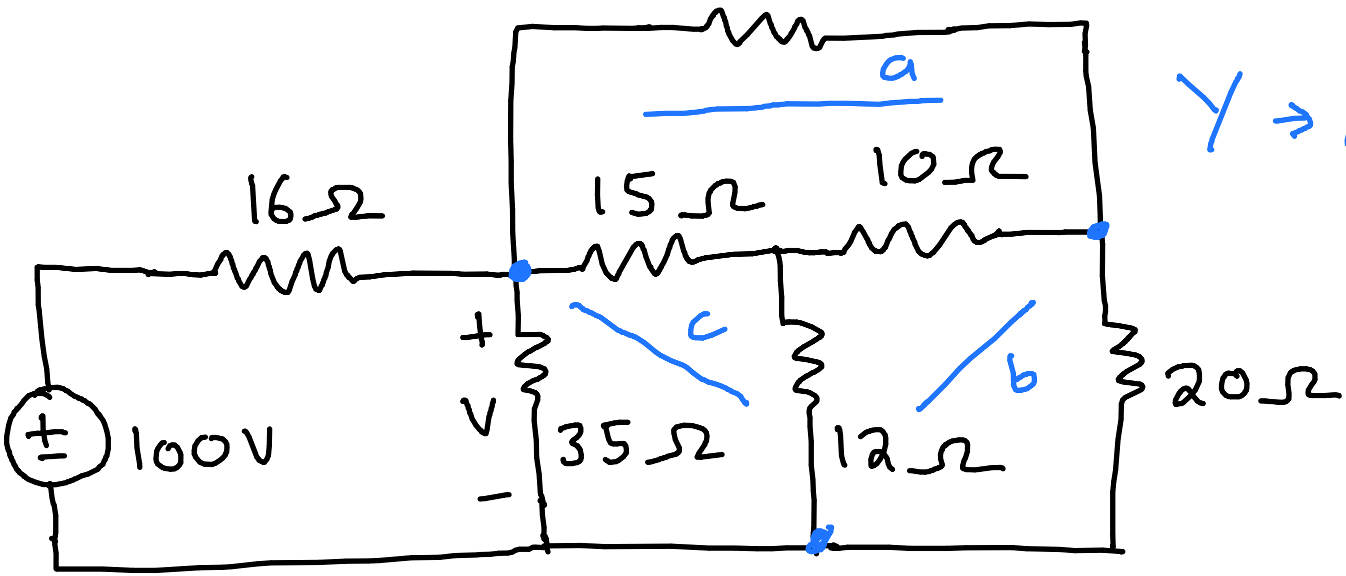
$$R_{eq} = 10 + 10 + 20 \| 40$$

$$= 20 + \frac{20 \times 40}{20 + 40}$$

$$= 33.333\ \Omega$$

2.56 Determine V in the circuit





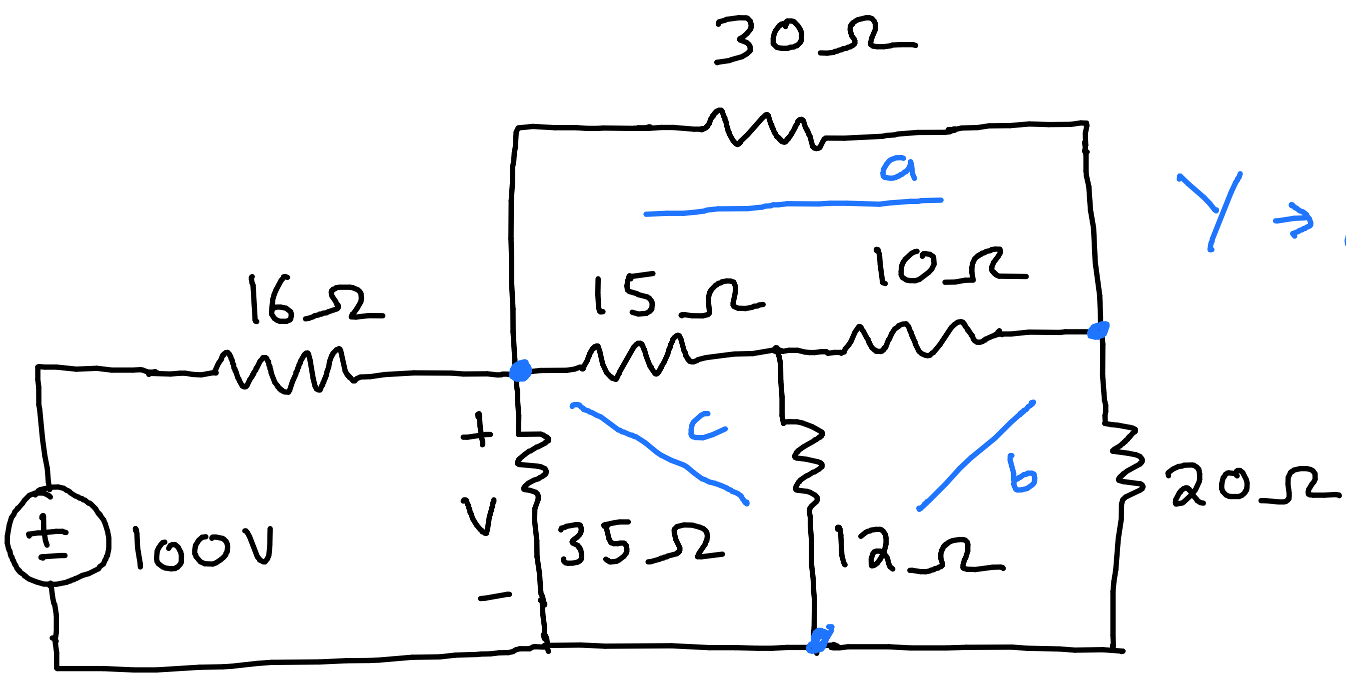
Y → Δ conversion

$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}$$

$R_1R_2 + R_2R_3 + R_3R_1$

$= 15\times10 + 15\times12 + 10\times12$

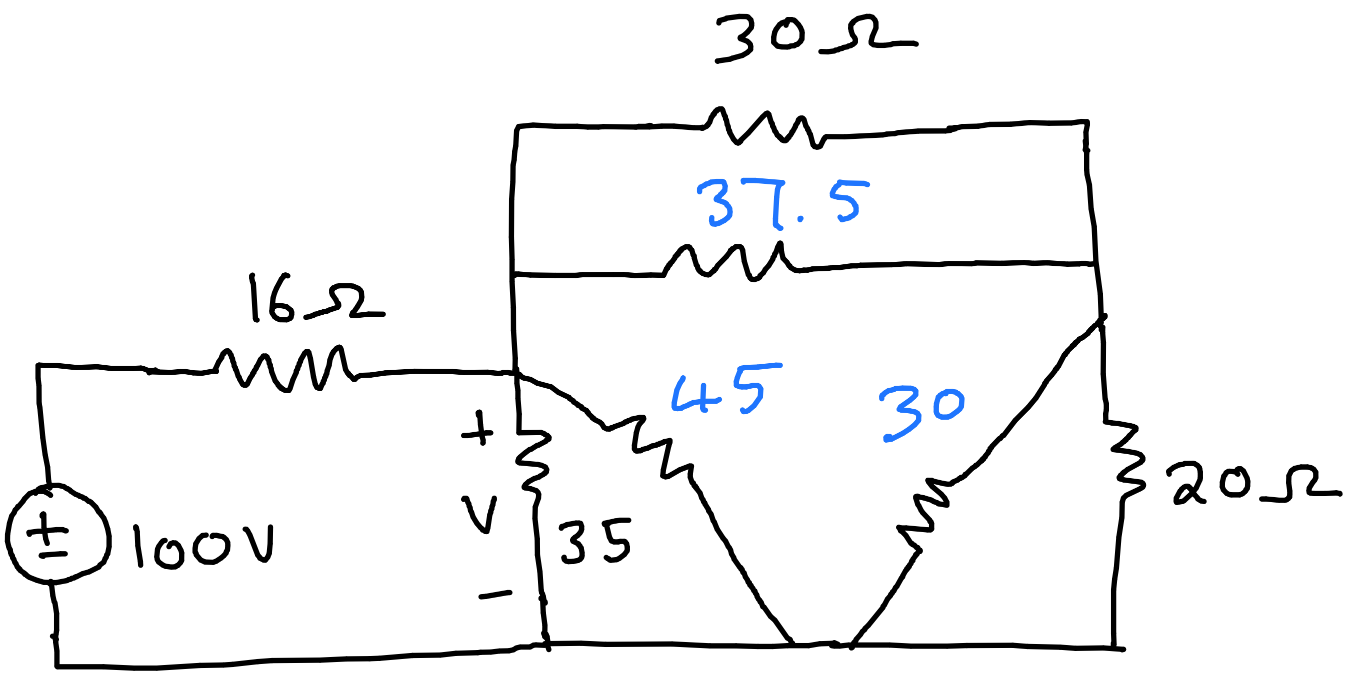
$= 450$



Y → Δ conversion

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

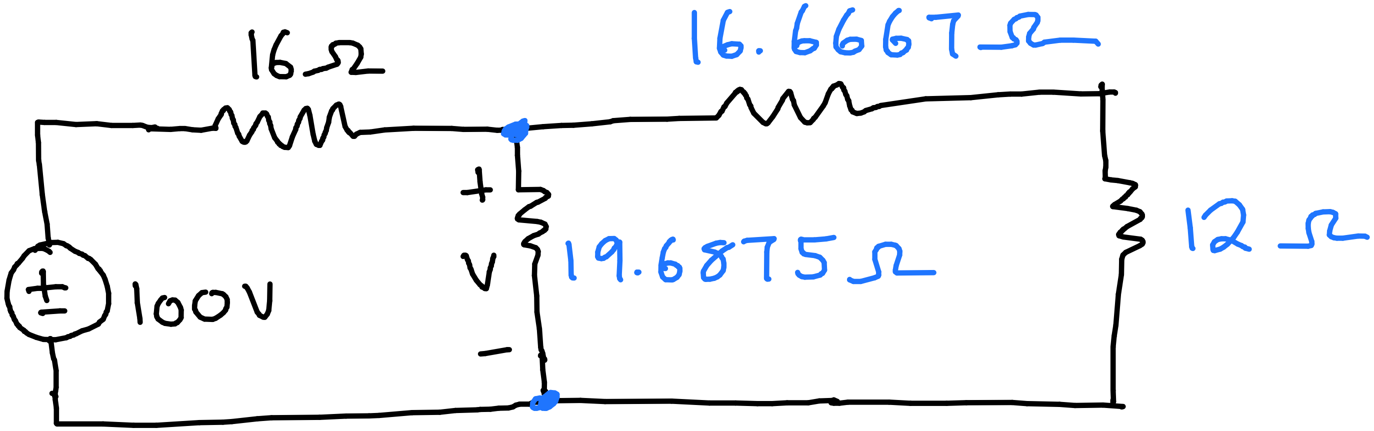
$$R_a = \frac{450}{12} = 37.5\,\Omega \qquad R_b = \frac{450}{15} = 30\,\Omega \qquad R_c = \frac{450}{10} = 45\,\Omega$$



$$35 \parallel 45 = \frac{35 \times 45}{35 + 45} = 19.6875\,\Omega$$

$$30 \parallel 37.5 = \frac{30 \times 37.5}{30 + 37.5} = 16.6667\,\Omega$$

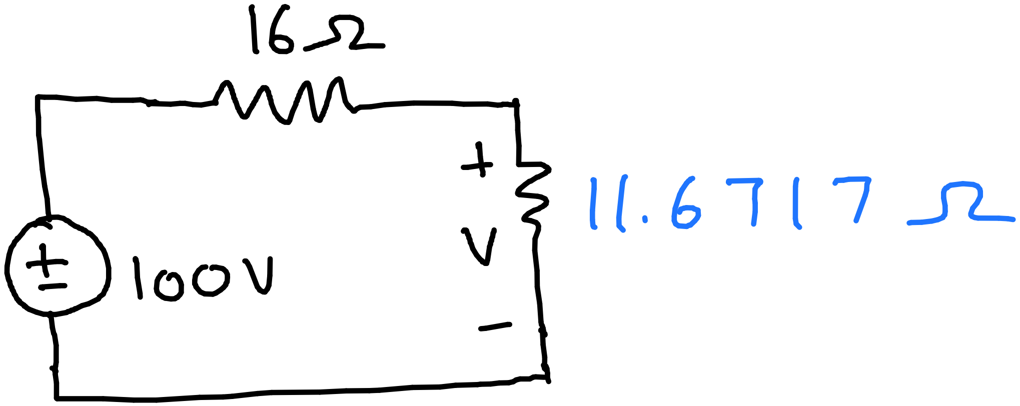
$$30 \parallel 20 = \frac{30 \times 20}{30 + 20} = 12\,\Omega$$



$$R_{eq} = (16.6667 + 12) \parallel 19.6875$$

$$= \frac{28.6667 \times 19.6875}{28.6667 + 19.6875}$$

$$= 11.6717\ \Omega$$



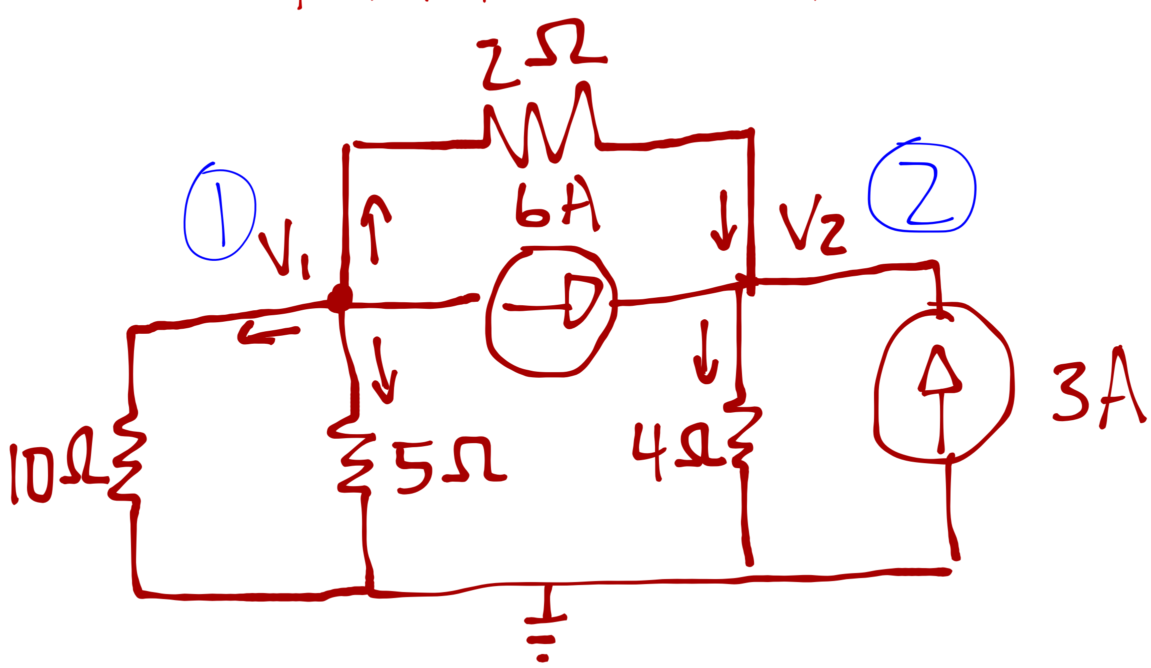
Voltage division

$$V_n = \frac{V R_n}{R_{eq}}$$

$$V = \frac{100 \times 11.6717}{16 + 11.6717}$$

$$= 42.179 \text{ V}$$

3.2 Find V_1 and V_2



@ Node ①: $\Sigma I_{in} = \Sigma I_{out}$

$$0 = \frac{V_1 - 0}{10} + \frac{V_1 - 0}{5} + \frac{V_1 - V_2}{2} + 6$$ → Multiply with 10

$$\therefore V_1 + 2V_1 + 5V_1 - 5V_2 = -60$$

↳ Multiply - out

$$-8V_1 + 5V_2 = 60$$

$$-8V_1 = 60 - 5V_2 \quad \ldots ①$$

@ Node ②: $\frac{V_1 - V_2}{2} + 3 + 6 = \frac{V_2}{4}$ → Multiply with 4

$$2V_1 - 2V_2 + (9 \times 4) = V_2$$ → Multiply with -4

$$4(-2V_1+3V_2=36)$$

$$-8V_1=144-12V_2 \ldots (2)$$

Solve using Simultaneous equations

(1) = (2)

$$60-5V_2=144-12V_2$$

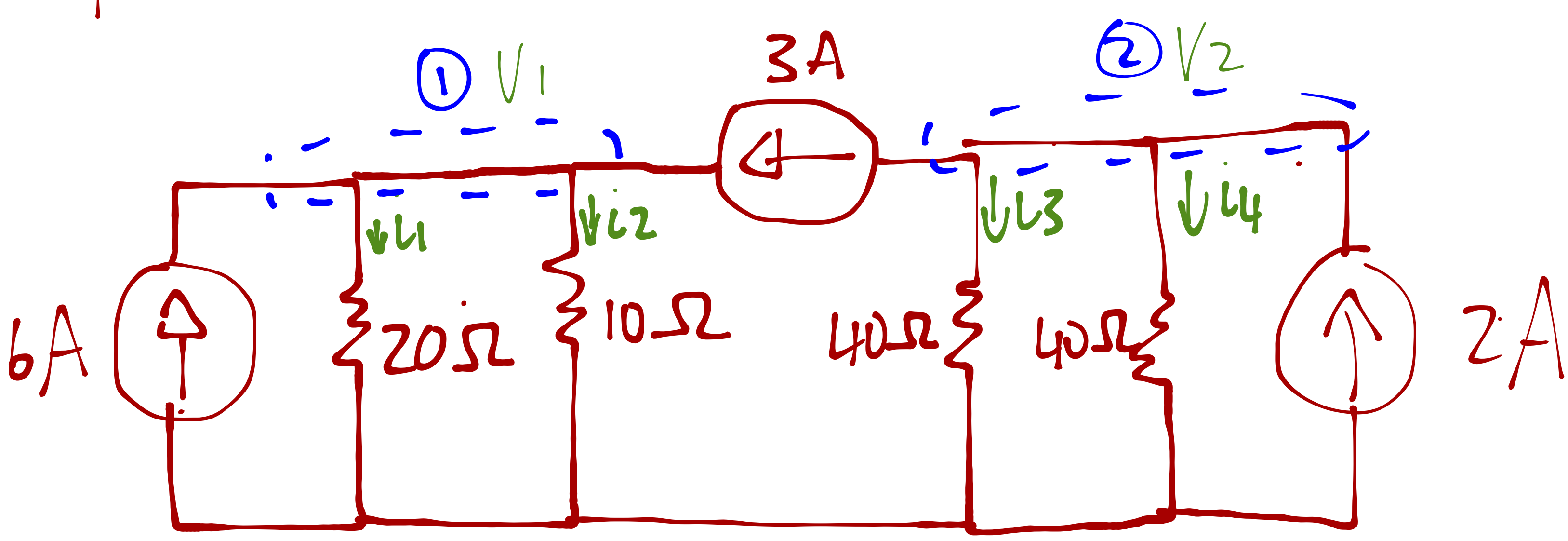
$$\therefore V_2=12V$$

Substitute V_2 into (1)

$$-8V_1=60-5(12)$$

$$\therefore V_1=0V$$

3.4 solve for i_1 to i_4



@Node ①: $\sum I_{in} = \sum I_{out}$

$$6+3 = \frac{V_1}{20} + \frac{V_1}{10}$$ → Multiply with 20

$$180 = V_1 + 2V_1$$

$$\therefore V_1 = 60V$$

$$i_1 = \frac{V_1}{20} = \frac{60}{20} = 3A \rightarrow$$

$$i_2 = \frac{V_1}{10} = \frac{60}{10} = 6A \rightarrow$$

@Node ②: $2 = 3 + i_3 + i_4$

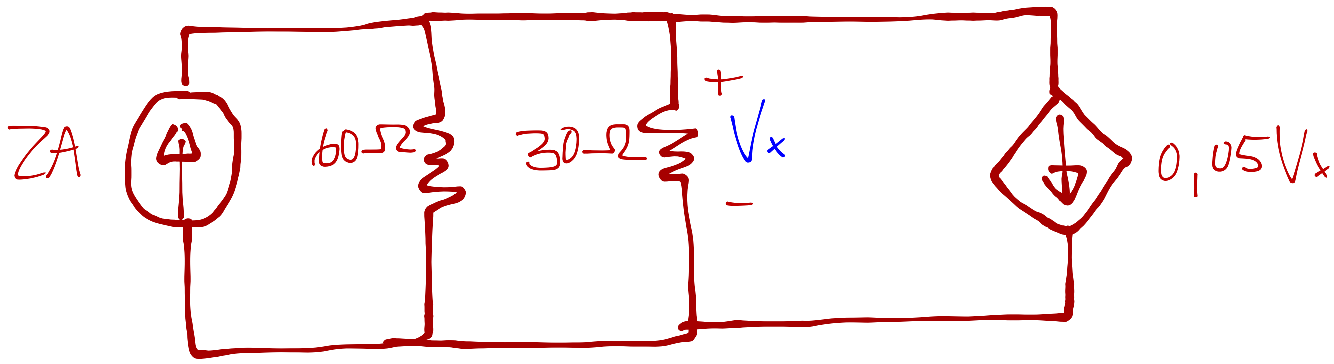
$$2-3 = \frac{V_2}{40} + \frac{V_2}{40}$$ → Multiply with 20

$$-20 = V_2$$

$$\therefore i_3 = i_4 = \frac{V_2}{40} = \frac{-20}{40}$$

$$= -0{,}5A \rightarrow$$

3.7 Solve for V_x



1 node with voltage = V_x

Nodal analysis: $\sum I_{in} = \sum I_{out}$

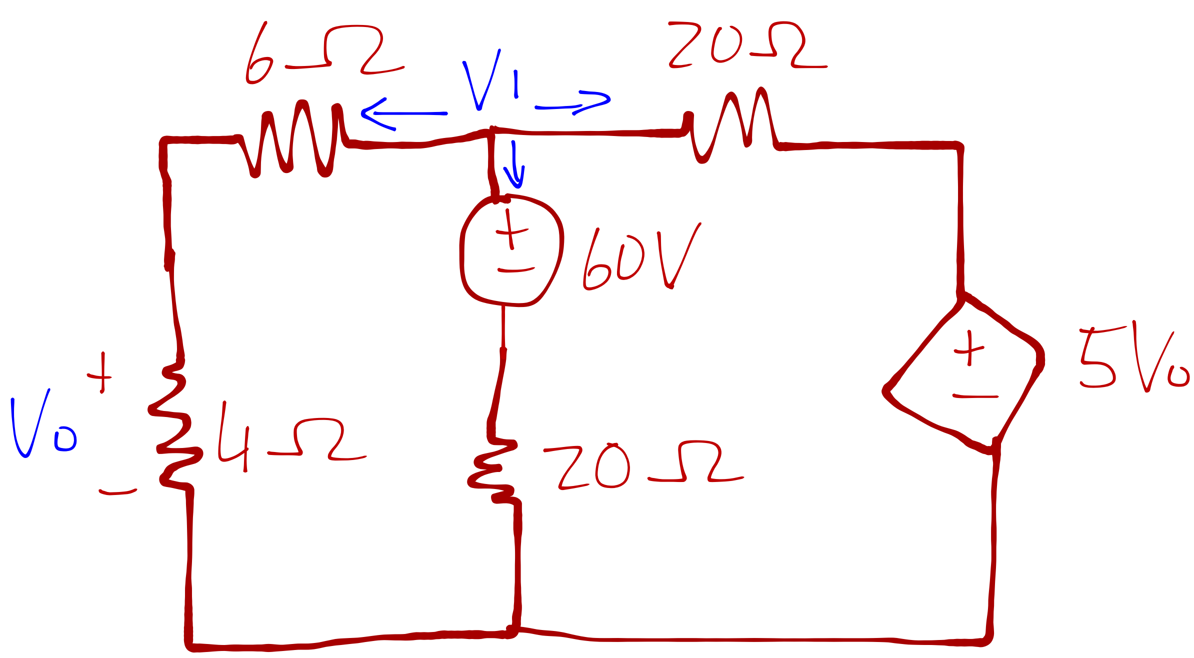
$$2 = \frac{V_x}{30} + \frac{V_x}{60} + 0{,}05V_x \rightarrow \text{Multiply with } 60$$

$$120 = V_x + 2V_x + 3V_x$$

$$120 = 6V_x$$

$$\therefore \underline{V_x = 20V}$$

3.8 Find V_o



Nodal analysis @ V_1

$$\sum I_{in} = \sum I_{out}$$

$$0 = \frac{V_1 - 0}{\underbrace{6+4}} + \frac{(V_1-60)-0}{20} + \frac{V_1 - 5V_0}{20} \rightarrow \text{Multiply with } 20$$

series resistances

$R_S = R_1 + R_2$

Used $I = \dfrac{V_{From} - V_{To}}{R}$

$$0 = 2V_1 + V_1 - 60 + V_1 - 5V_0$$

$$60 = 4V_1 - 5V_0 \quad \ldots \; 1$$

Voltage division:

$$V_0 = \left(\frac{4}{R_S}\right) V_1$$

$$V_0 = \left(\frac{4}{4+6}\right) V_1$$

$$= \frac{2}{5} V_1 \quad \ldots \; 2$$

$\therefore$ Substitute 2 in 1

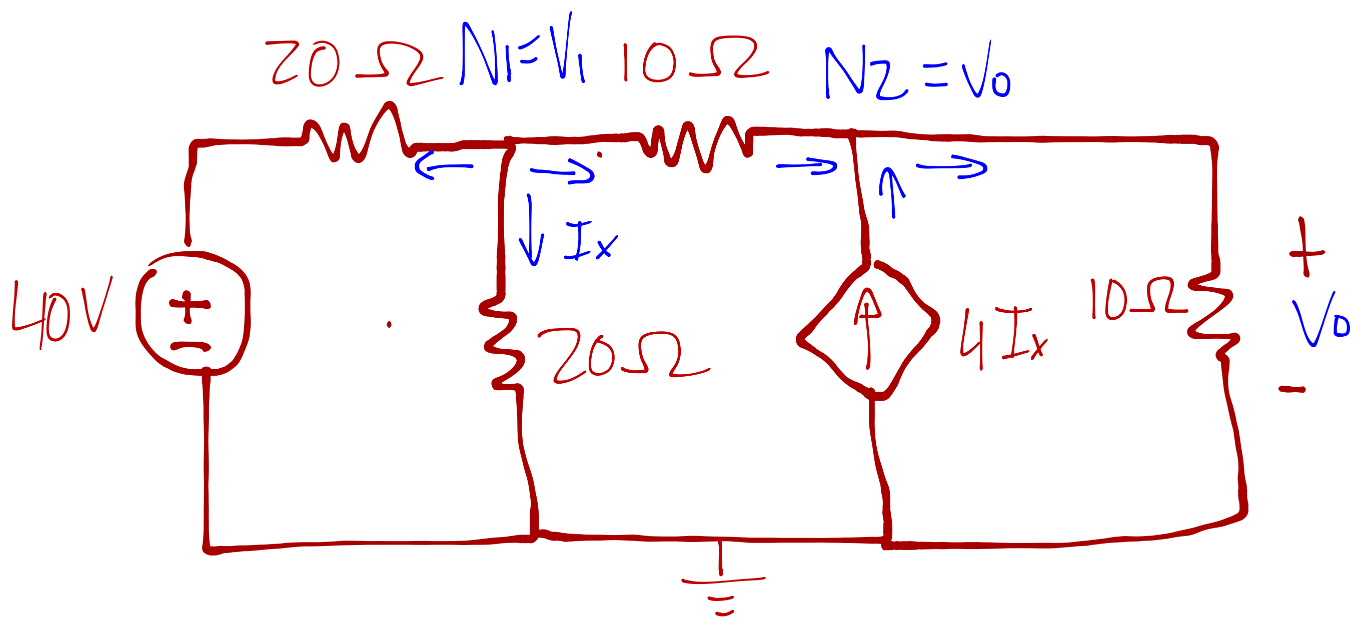
$$\Rightarrow 4V_1 - 5\left(\frac{2}{5}V_1\right) = 60$$

$$V_1 = 30V$$

Substitute V_1 in 2

$$\therefore V_0 = \frac{2}{5}(30) = \underline{12V}$$

3.12 Determine V_o



@ Node 1: $\Sigma I_{in} = \Sigma I_{out}$

$$0 = \frac{V_1 - 40}{20} + \frac{V_1 - 0}{20} + \frac{V_1 - V_o}{10}$$ → Multiply with 20

$$0 = V_1 - 40 + V_1 + 2V_1 - 2V_o$$

$$40 = 4V_1 - 2V_o \quad \ldots (1)$$

@ Node 2: $$4I_x + \frac{V_1 - V_o}{10} = \frac{V_o - 0}{10}$$ → Multiply with 10

$$40I_x + V_1 = V_o + V_o \quad \ldots (2)$$

Ohms law: $I = \frac{V}{R}$

$$\therefore I_x = \frac{V_1}{20} \quad \ldots (3)$$

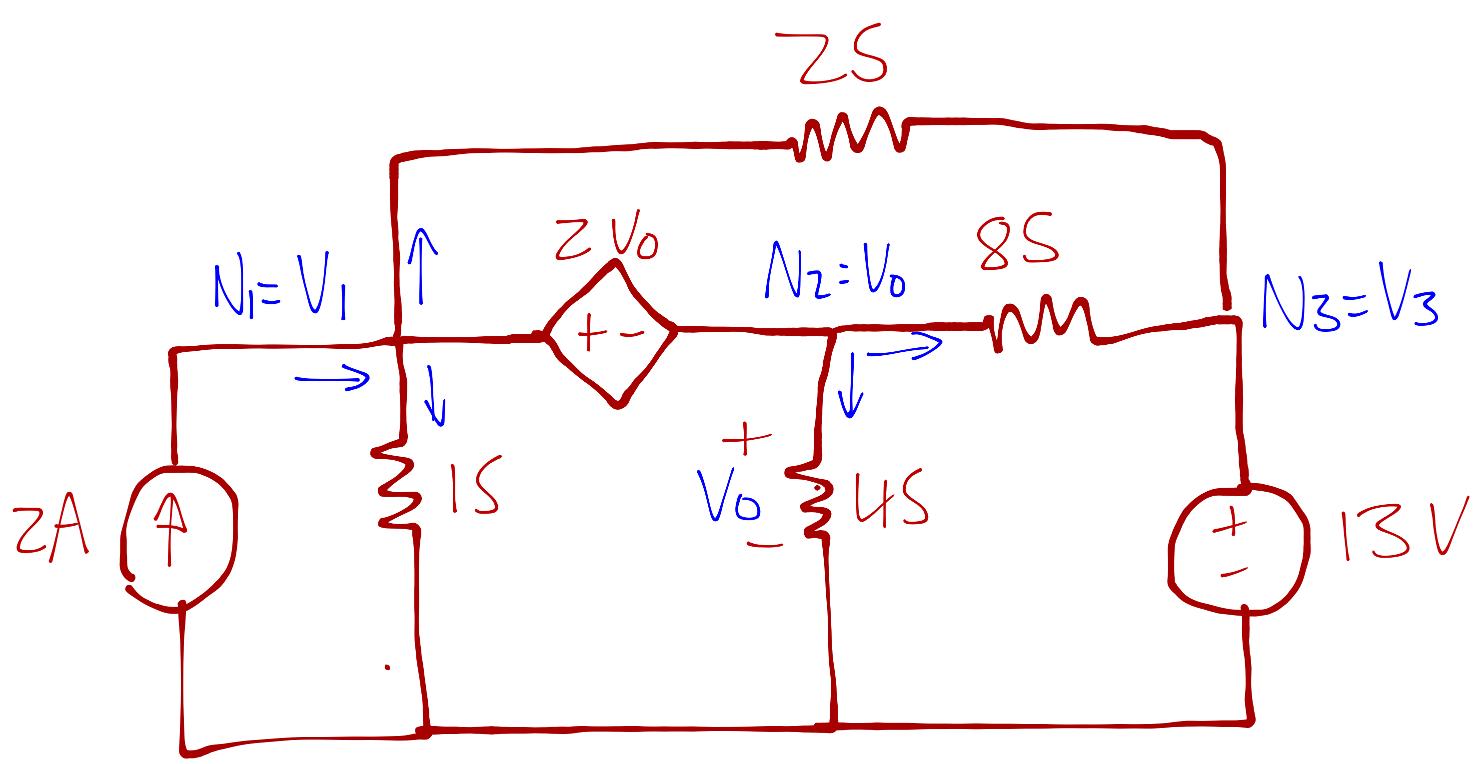
Substitute (3) into (2)

$$2V_1 + V_1 = 2V_0$$
$$3V_1 = 2V_0$$
$$\therefore V_1 = \frac{2}{3}V_0 \quad \ldots \quad (4)$$

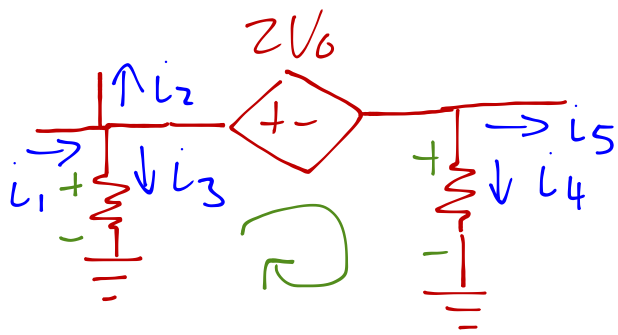
Substitute (4) into (1)

$$40 = 4\left(\frac{2}{3}V_0\right) - 2V_0$$
$$\therefore \underline{V_0 = 60}$$

3.16 Find V_1 to V_3



See the conjunction of N1 and N2 as a supernode



Note !!!

$$\boxed{\frac{1}{S} = R}$$

KCL as a single node

$$\Sigma I_{in} = \Sigma I_{out}$$

$$2 = 1(V_1 - 0) + 2(V_1 - V_3) + 4(V_0 - 0) + 8(V_0 - V_3)$$

$$2 = 3V_1 + 12V_0 - 10V_3 \quad \ldots \ (1)$$

KVL @ supernode

$$\Sigma V = 0$$

$$-V_1 + 2V_0 + V_0 = 0$$

$$\therefore V_1 = 3V_0 \quad \ldots \ (2)$$

Note $V_3 = 13V$ ... (3)

Substitute ③ and ② into ①

$$2 = 3V_1 + 12\left(\frac{1}{3}V_1\right) - 10(13)$$

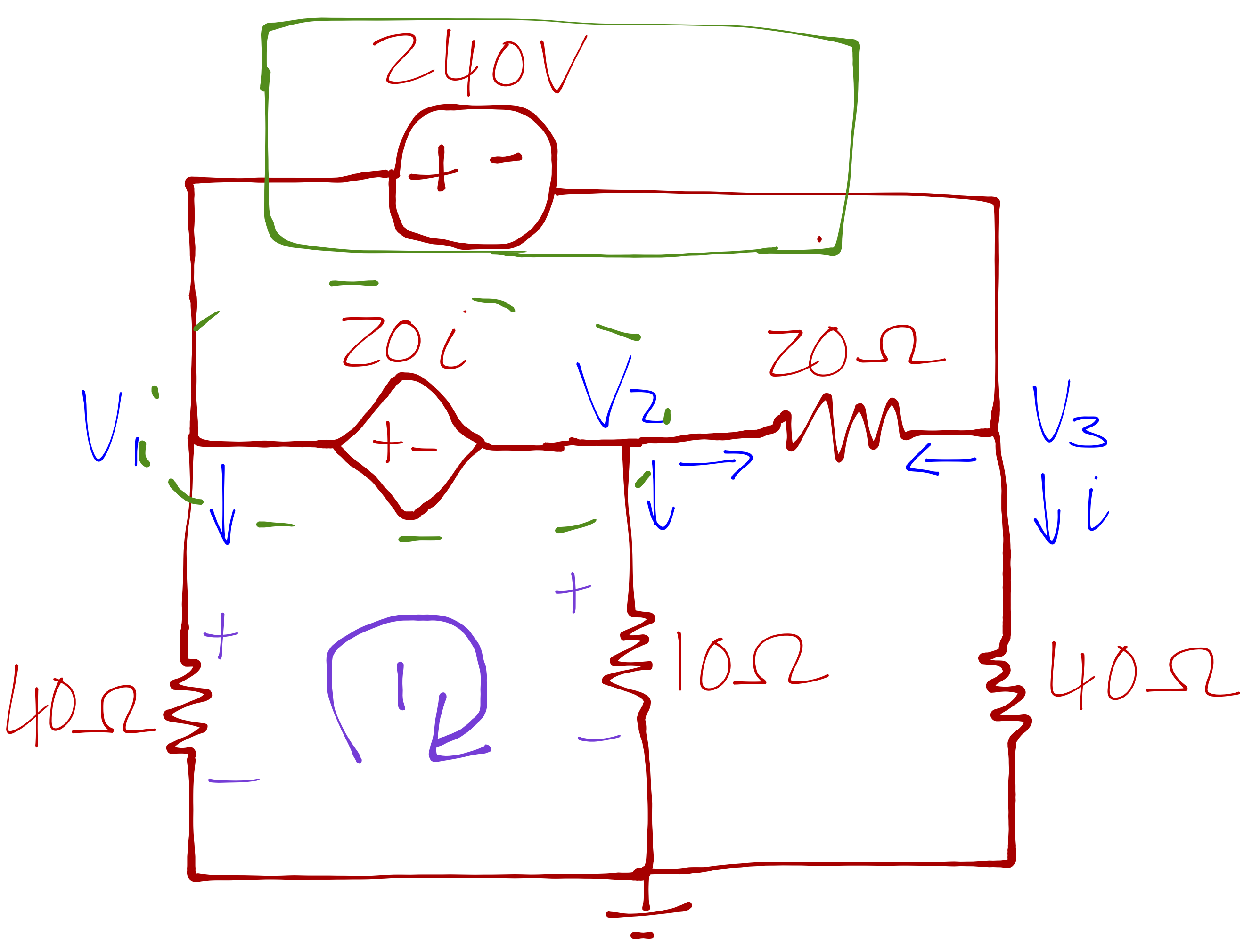
$$\therefore V_1 = 18.857\,V$$

Substitute V_1 into ②

$$\frac{1}{3}(18.857) = V_0$$

$$\therefore V_0 = V_2 = 6.286\,V$$

3.20 Find V_1 to V_3



See the conjunction of Node 1 and Node 2

as a supernode

Note between Node 1 and Node 3 is another supernode. For KCL we see Node 1, 2 and 3 as one node.

$$\Sigma I_{in} = \Sigma I_{out}$$

$$0 = \frac{V_1 - 0}{40} + \frac{V_2 - 0}{10} + \frac{V_3 - 0}{40} + \frac{V_2 - \cancel{V_3}}{\cancel{20}} + \frac{V_3 - \cancel{V_2}}{\cancel{20}}$$

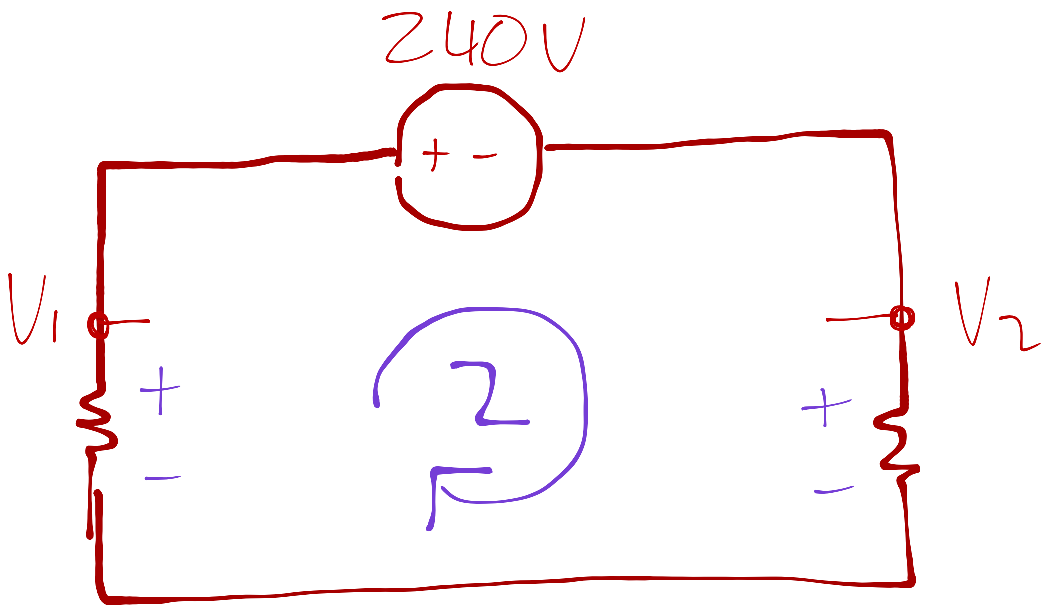
Multiply with 40

$$0 = V_1 + 4V_2 + V_3 \quad \ldots \text{(1)}$$

KVL1: $\Sigma V = 0$

$$-V_1 + 20i + V_2 = 0$$

$$\therefore V_1 - V_2 = 20i \quad \ldots \text{(2)}$$



KVL2: $-V_1 + 240 + V_3 = 0$

$$\therefore V_1 - V_3 = 240 \quad \ldots (3)$$

Ohms law $i = \frac{V}{R}$

$$\therefore i = \frac{V_3}{40} \quad \ldots (4)$$

Substitute (4) into (3)

$$V_1 - 40i = 240$$

$$V_1 - 240 = 40i$$

$$\frac{1}{2}(V_1 - 240) = 20i \quad \ldots (5)$$

Set 5 = 2

$$V_1 - V_2 = \frac{1}{2}(V_1 - 240)$$

$$V_2 = V_1 - \frac{1}{2}V_1 + 120 \quad \ldots 6$$

Substitute 6 and 3 into 1

$$V_1 + 4\left(\frac{1}{2}V_1 + 120\right) + (V_1 - 240) = 0$$

$$4V_1 = -240$$

$$\underline{V_1 = -60\ V}$$

Substitute V_1 into 6 and 3

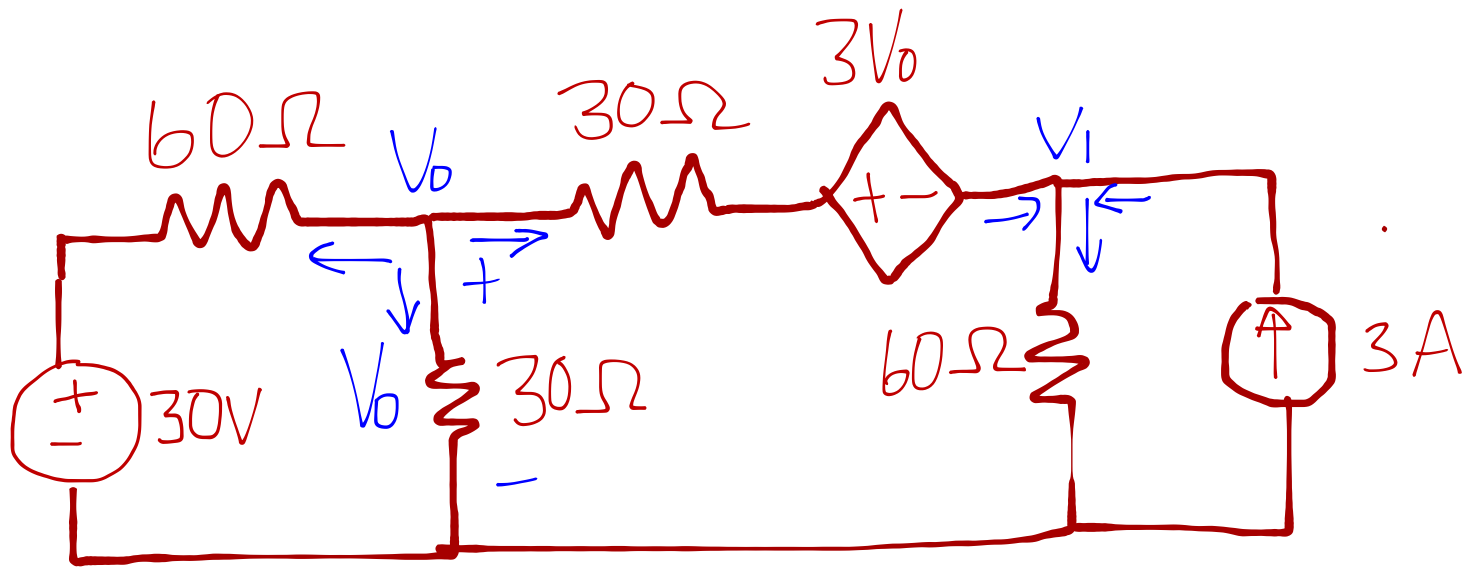
3: $-60 - 240 = V_3$

$$\therefore \underline{V_3 = -300\ V}$$

6: $\frac{1}{2}(-60) + 120 = V_2$

$$\therefore \underline{V_2 = 90\ V}$$

3.23 Find V_o



@Node 0: $\sum I_{in} = \sum I_{out}$

$$0 = \frac{V_o - 30}{60} + \frac{V_o - 0}{30} + \frac{V_o - (3V_o + V_1)}{30}$$

→ Multiply with 60

$$0 = V_o - 30 + 2V_o + 2V_o - 6V_o - 2V_1$$

$$30 = -V_o - 2V_1$$

$$\therefore V_o = -30 - 2V_1 \quad \ldots (1)$$

@Node 1: $\frac{V_o - (3V_o + V_1)}{30} + 3 = \frac{V_1}{60}$ → Multiply with 60

$$2V_o - 6V_o - 2V_1 + 180 = V_1$$

$$-4V_o - 3V_1 = -180 \quad \ldots (2)$$

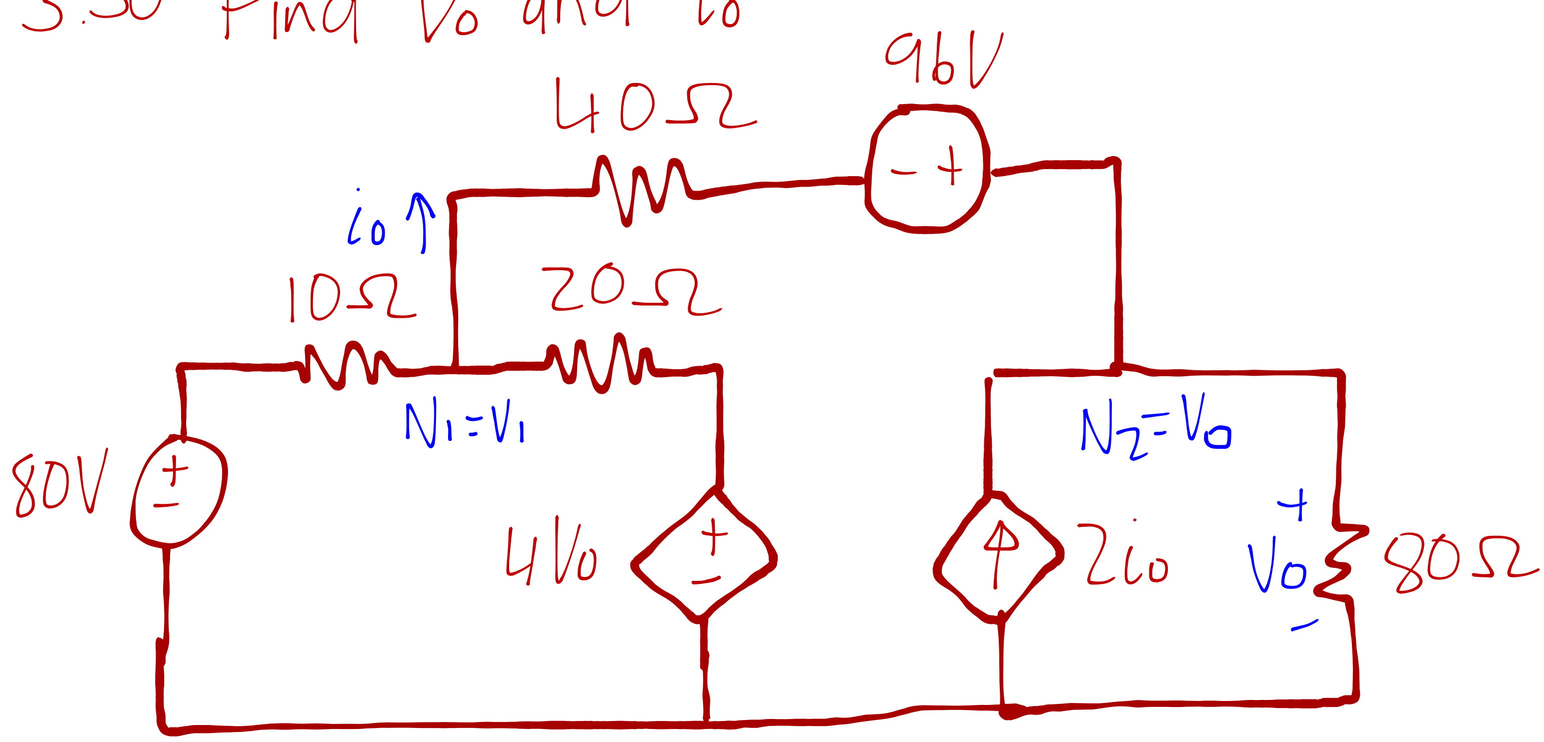
Substitute ① in ②

$$-4(-30-2V_1)-3V_1=-180$$
$$120+8V_1-3V_1=-180$$
$$5V_1=-300$$
$$V_1=-60\,V$$

Substitute V_1 into ①

$$\therefore V_0=-30-2(60)$$
$$=\underline{90\,V}$$

3.30 Find V_o and i_o



@ Node 1: $\Sigma I_{in} = \Sigma I_{out}$

$$\frac{80-V_1}{10} + \frac{4V_0 - V_1}{20} = \frac{V_1 - (V_0 - 96)}{40}$$ → Multiply with 40

$$320 - 4V_1 + 8V_0 - 2V_1 = V_1 - V_0 + 96$$

$$224 = 7V_1 - 9V_0 \quad \ldots \text{①}$$

@ Node 2: $i_0 + 2i_0 = \frac{V_0 - 0}{80} \quad \ldots \text{Ⓐ}$

Replace i_0 with $\frac{V_1 - (V_0 - 96)}{40}$

$$\therefore \quad 3\left(\frac{V_1 + 96 - V_0}{40}\right) = \frac{V_0}{80}$$

$$6V_1 + 576 - 6V_0 = V_0$$

$$6V_1 - 7V_0 = -576 \quad \ldots \text{②}$$

Use Cramers rule to solve equation ② and ③ simultaneously

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_0 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 7 & -9 \\ 6 & -7 \end{bmatrix} \begin{bmatrix} V_1 \\ V_0 \end{bmatrix} = \begin{bmatrix} 224 \\ -576 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 7 & -9 \\ 6 & -7 \end{vmatrix} = (7)(-7) - (-9)(6) = -49 + 54$$

$$\Delta = 5$$

replace $\hat{y}$ into 1st column

$$\Delta_1 = \begin{vmatrix} 224 & -9 \\ -576 & -7 \end{vmatrix} = 224(-7) - (-9)(-576)$$

$$= -1568 - 5184$$

$$= -6752$$

$$\Delta_2 = \begin{vmatrix} 7 & 224 \\ 6 & -576 \end{vmatrix} = 7(-576) - (224)(6)$$

$$= -4032 - 1344$$

$$= -5376$$

using determinants V_1 and V_o can be calculated

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{-6752}{5} = -1350{,}4\,V$$

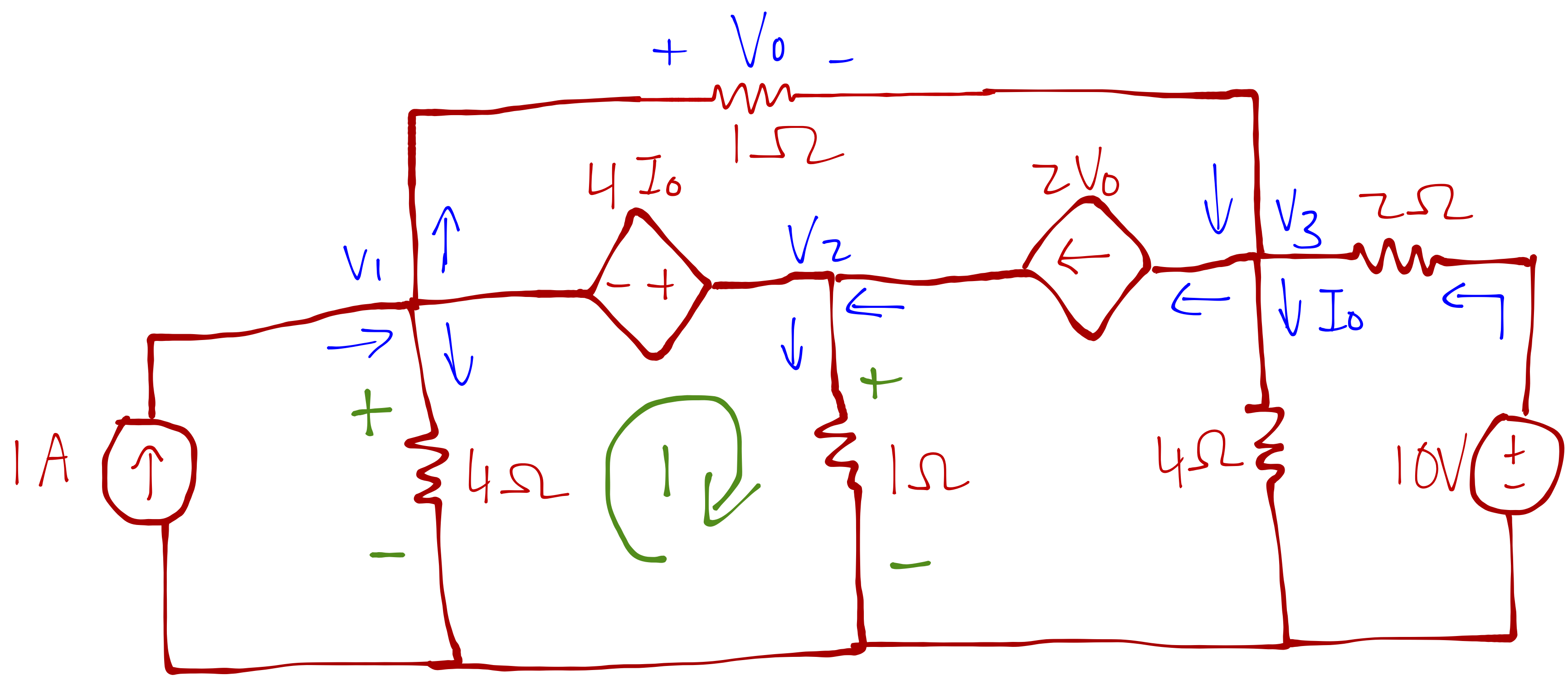
$$V_o = \frac{\Delta_2}{\Delta} = \frac{-2688}{5} = -1075{,}2\,V$$

substitute V_o into A

$$3 i_o = \frac{-1075.2}{80}$$

$$\therefore i_o = -4{,}48\,A$$

3.31 Find V_1 to V_3



Start by finding equations from the Supernode between Node 1 and 2

KCL: $\Sigma I_{in} = \Sigma I_{out}$

$$1 + 2V_o = \frac{V_1 - 0}{4} + \frac{V_1 - V_3}{1} + \frac{V_2 - 0}{1} \quad \ldots \text{(1)}$$

Note V_o is between V_1 and V_3

$$\therefore V_o = V_1 - V_3 \quad \ldots \text{(2)}$$

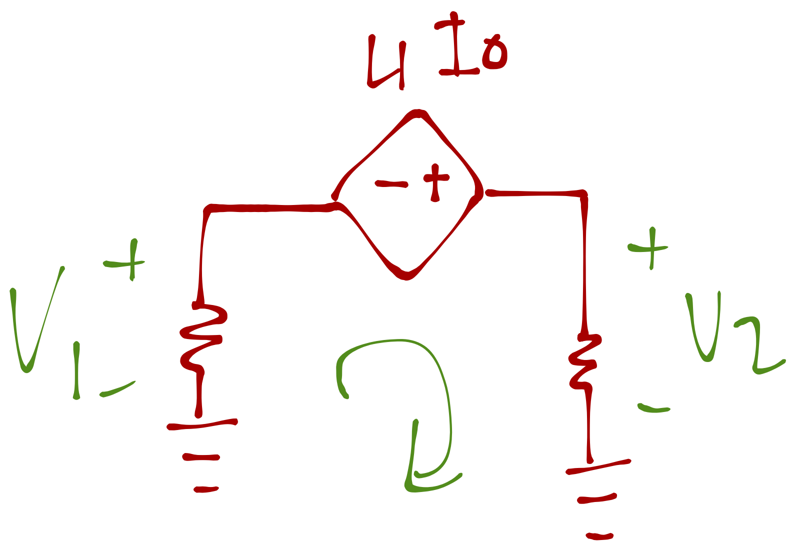
Substitute (2) into (1)

$$1+2(V_1-V_3)=\frac{V_1}{4}+V_1-V_3+V_2 \rightarrow \text{Multiply with } 4$$

$$4+8V_1-8V_3=V_1+4V_1-4V_3+4V_2$$

$$4=-3V_1+4V_2+4V_3 \quad \ldots \text{③}$$

KVL1:



$$\Rightarrow \sum V=0$$

$$-V_1-4I_o+V_2=0$$

$$\therefore V_2=V_1+4I_o$$

Note using Ohms law $i_o=\frac{V_3}{4} \quad \ldots \text{④}$

$$\therefore V_2=V_1+V_3 \quad \ldots \text{⑤}$$

@ Node 3: $\Sigma I_{in} = \Sigma I_{out}$

$$\frac{V_1 - V_3}{1} + \frac{10 - V_3}{2} = 2V_0 + \frac{V_3 - 0}{4}$$ → Multiply with 4

$$4V_1 - 4V_3 + 20 - 2V_3 = 8V_0 + V_3$$

Substitute 2 into equation

$$4V_1 - 7V_3 + 20 = 8(V_1 - V_3)$$

$$V_3 = 4V_1 - 20 \quad \ldots 6$$

Substitute 5 into 3

$$4 = -3V_1 + 4(V_1 + V_3) + 4V_3 \quad \ldots 7$$

Substitute 6 into 7

$$4 = V_1 + 4(4V_1 - 20) + 4(4V_1 - 20)$$

$$164 = 33V_1$$

$$\therefore V_1 = 4{,}97\,V$$

Substitute V_1 into (6)

$$V_3 = 4(4{,}97) - 20$$

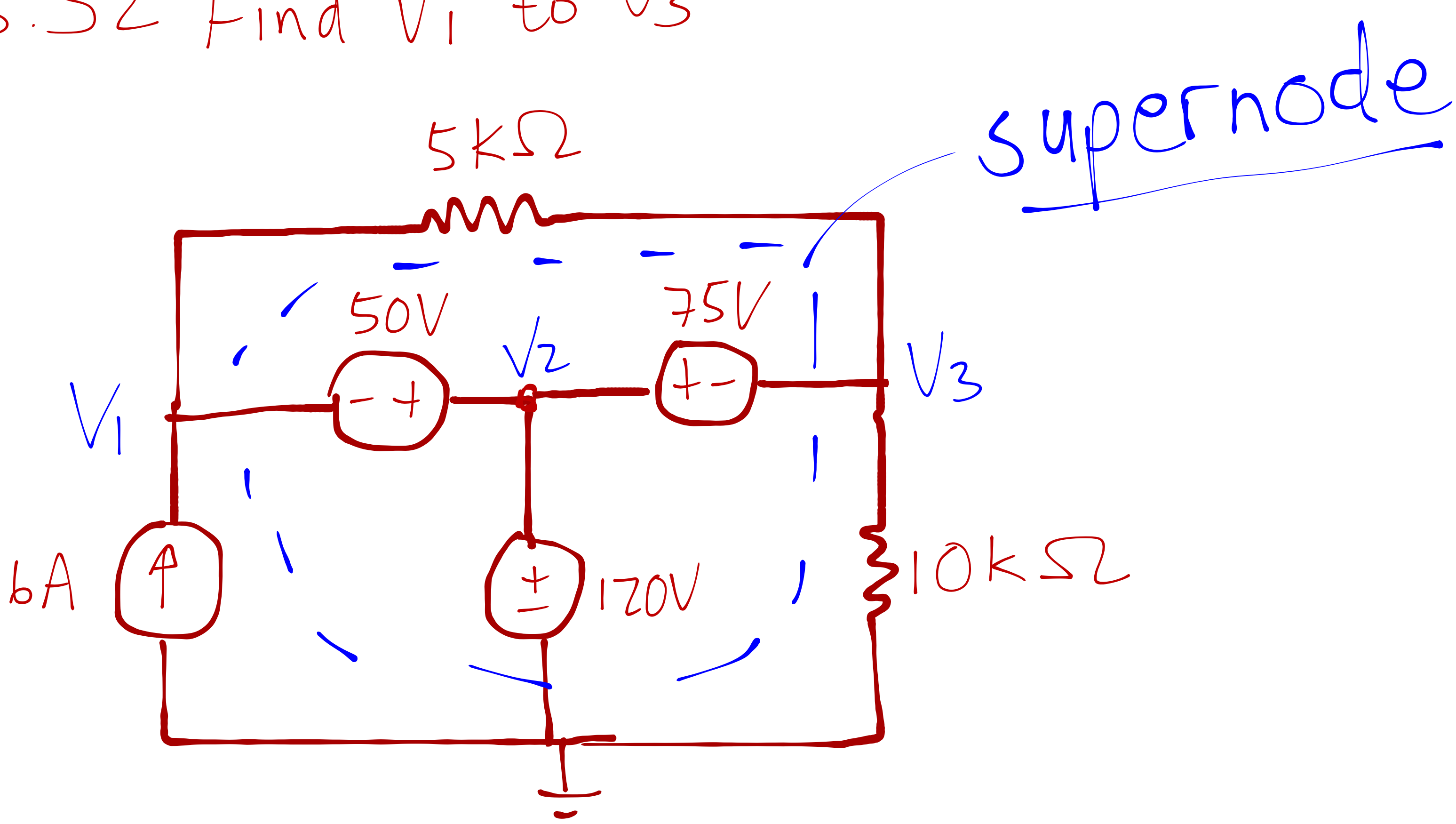
$$\therefore V_3 = -0{,}12\,V$$

Substitute V_1 and V_3 into (5)

$$V_2 = 4{,}97 - 0{,}12$$

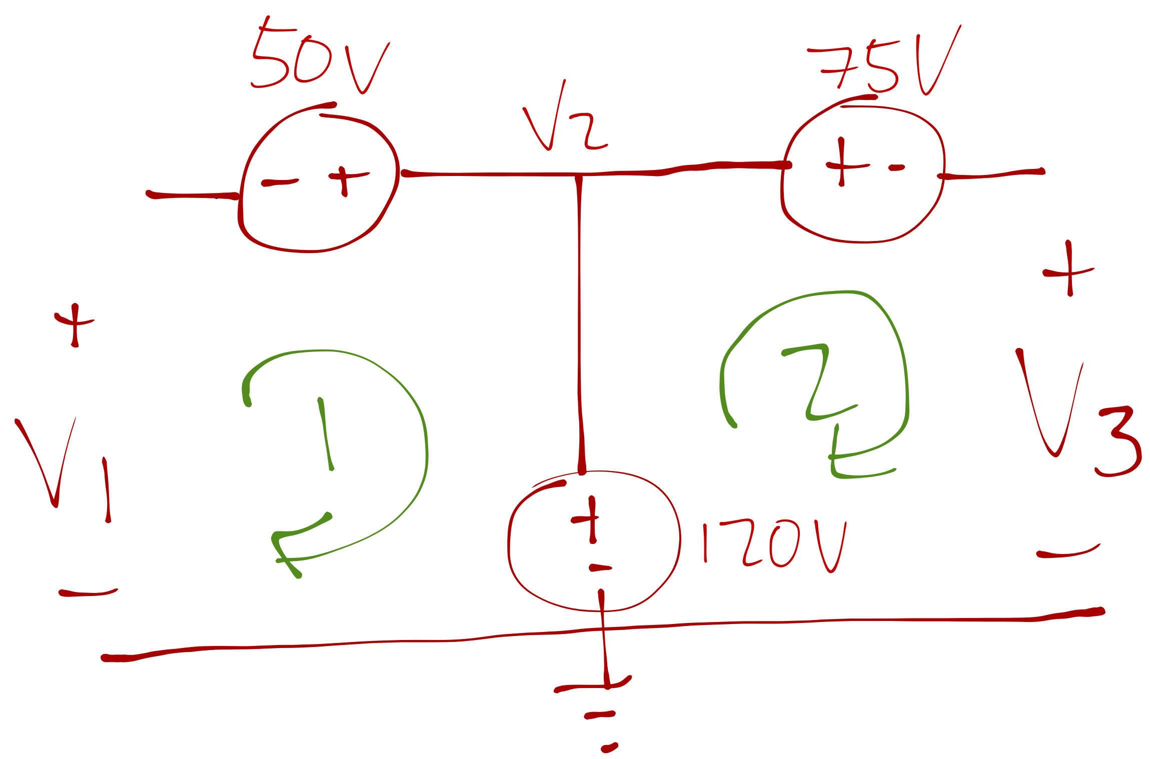
$$\therefore V_2 = 4{,}85\,V$$

3.32 Find V_1 to V_3



The middle section of the circuit forms one big supernode

The supernode has 2 independant loops which both could be used with KVL for equations



KVL ①: $\Sigma V = 0$

$$-V_1 - 50V + 120V = 0$$

$$\therefore \underline{V_1 = 70V}$$

Note $V_2 = 120V$ by visual inspection

KVL 2 : $-120 + 75 + V_3 = 0$

$$V_3 = 45V$$